



Analog IC Design Implementation and Verification
Weeks 3 and 4 – Current Mirrors and Amplifiers

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Introduction: Current Mirrors

- Characterization of current mirrors
- Influences on accuracy
 - Channel modulation parameter (λ)
 - Transistor mismatch
 - Transistor aspect ratios
- Types of current mirrors
 - Basic MOS
 - Cascode MOS
 - Large output swing cascode MOS
 - Self-biased cascode MOS
 - Wilson MOS
 - Regulated cascode MOS



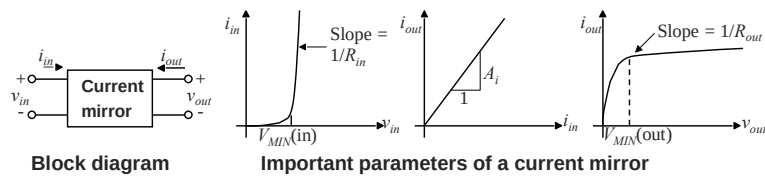
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Characterization of Current Mirrors

A current mirror is essentially a current amplifier. The ideal characteristics of a current amplifier are:

- Output current is linearly related to the input current, $i_{out} = A_i i_{in}$.
- Input resistance is zero.
- Output resistance is infinity.
- Bandwidth is infinite.
- No added noise.
- $V_{MIN}(in)$ is the range of v_{in} over which the input resistance is not small.
- $V_{MIN}(out)$ is the range of v_{out} over which the output resistance is not large.

Therefore, R_{out} , R_{in} , $V_{MIN}(out)$, $V_{MIN}(in)$, and A_i characterize the current mirror:



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Basic MOS Current Mirror

Assume that $v_{DS2} > v_{GS} - V_{T2}$, then:

$$\frac{i_O}{i_I} = \left(\frac{L_1 W_2}{W_1 L_2} \right) \left(\frac{V_{GS} - V_{T1}}{V_{GS} - V_{T2}} \right)^2 \left[\frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}} \left(\frac{k_2'}{k_1'} \right) \right]$$

If the transistors are matched, then $k_1' = k_2'$ and $V_{T1} = V_{T2}$ to give:

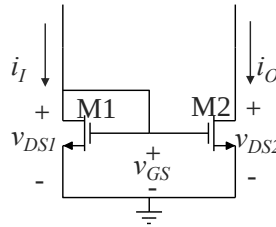
$$\frac{i_O}{i_I} = \left(\frac{L_1 W_2}{W_1 L_2} \right) \left(\frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}} \right)$$

If $v_{DS1} = v_{DS2}$, then:

$$\frac{i_O}{i_I} = \left(\frac{L_1 W_2}{W_1 L_2} \right)$$

Therefore, the sources of error are:

- $v_{DS1} \neq v_{DS2}$
- M1 and M2 not matched



Basic MOS current mirror

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Influence of the Channel Modulation Parameter λ

If the transistors are matched and the W/L ratios are equal, then:

$$\frac{i_O}{i_I} = \frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}}$$

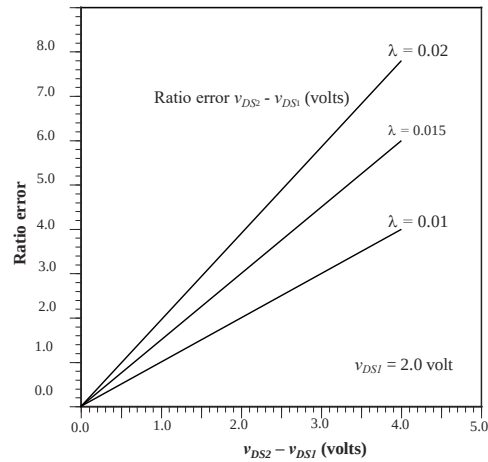
if the channel modulation parameter is the same for both transistors ($L_1 = L_2$).

The ratio error, e, versus drain voltage difference:

$$e = \left[\frac{1 + \lambda v_{DS2}}{1 + \lambda v_{DS1}} - 1 \right] \times 100\%$$

Notice that one can use this effect to measure λ :

Measure V_{DS1} , V_{DS2} , i_I and i_O and solve the previous equation for the channel modulation parameter λ .



Ratio error for different values of λ

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Influence of Mismatched Transistors

Assume that $v_{DS1} = v_{DS2}$ and that $k_1' \neq k_2'$ and $V_{T1} \neq V_{T2}$. Therefore:

$$\frac{i_O}{i_I} = \frac{k_2' (v_{GS} - v_{T2})^2}{k_1' (v_{GS} - v_{T1})^2}$$

How do you analyze the mismatch? Use plus and minus worst-case approach. Define:

$$\Delta k' = k_2' - k_1' \text{ and } k' = 0.5(k_2' + k_1') \Rightarrow k_1' = k' - 0.5\Delta k' \text{ and } k_2' = k' + 0.5\Delta k'$$

$$\Delta V_T = V_{T2} - V_{T1} \text{ and } V_T = 0.5(V_{T1} + V_{T2}) \Rightarrow V_{T1} = V_T - 0.5\Delta V_T \text{ and } V_{T2} = V_T + 0.5\Delta V_T$$

Substituting these terms in the previous equation gives:

$$\frac{i_O}{i_I} = \frac{k' + 0.5\Delta k' (v_{GS} - V_T - 0.5\Delta V_T)^2}{k' - 0.5\Delta k' (v_{GS} - V_T + 0.5\Delta V_T)^2} = \frac{\left(1 + \frac{\Delta k'}{2k'}\right) \left(1 - \frac{\Delta V_T}{2(v_{GS} - V_T)}\right)^2}{\left(1 - \frac{\Delta k'}{2k'}\right) \left(1 + \frac{\Delta V_T}{2(v_{GS} - V_T)}\right)^2}$$

Assuming that the terms added to or subtracted from 1 are smaller than unity gives:

$$\frac{i_O}{i_I} = \left(1 + \frac{\Delta k'}{2k'}\right) \left(1 + \frac{\Delta k'}{2k'}\right) \left(1 - \frac{\Delta V_T}{2(v_{GS} - V_T)}\right)^2 \left(1 - \frac{\Delta V_T}{2(v_{GS} - V_T)}\right)^2 \approx 1 + \frac{\Delta k'}{k'} - \frac{2\Delta V_T}{v_{GS} - V_T}$$

Assuming $\Delta k' / k' = \pm 5\%$ and $\Delta V_T / (v_{GS} - V_T) = \pm 10\%$,

$$\therefore \frac{i_O}{i_I} \approx 1 \pm 0.05 \pm (-0.20) = 1 \pm 0.25 \Rightarrow \pm 15\% \quad (\text{If tolerances are correlated})$$

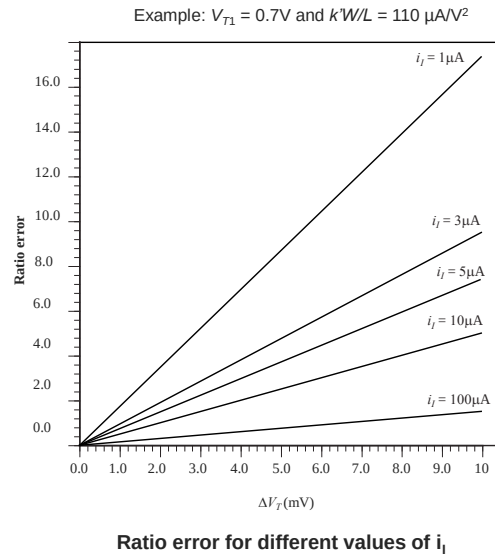


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Illustration: Offset Voltage Error Influence

- Large $V_{GS} - V_T$ reduces sensitivity to V_T errors.
- Larger L reduces the error in two ways:
 - Improves $W \cdot L$ to lower V_T mismatch.
 - Reduces g_m/i_d to lower the current sensitivity to the V_T mismatch.

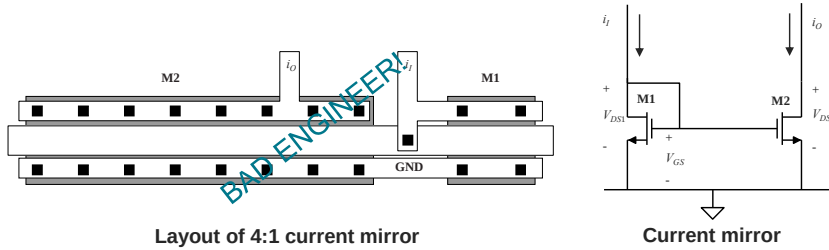
Note: There is a problem because large L increases V_{dsat} and a large area slows the circuit.



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Layout of Current Mirrors

The figure shows the layout of a one-to-four current amplifier. Assume that the lengths are identical ($L_1 = L_2$) and find the ratio error if $W_1 = 5 \pm 0.1 \mu\text{m}$. The actual widths of the two transistors are $W_1 = 5 \pm 0.1 \mu\text{m}$ and $W_2 = 20 \pm 0.1 \mu\text{m}$.



Reason: Notice that the tolerance is not multiplied by the nominal gain factor of 4. The ratio of W_2 to W_1 and consequently the gain of the current amplifier is:

$$\frac{i_o}{i_i} = \frac{W_2}{W_1} = \frac{20 \pm 0.1}{5 \pm 0.1} = 4 \left(\frac{1 \pm (0.1/20)}{1 \pm (0.1/5)} \right) \approx 4 \left(1 \pm \frac{0.1}{20} \right) \left(1 - \frac{\pm 0.1}{5} \right) \approx 4 \left(1 \pm \frac{0.1}{20} - \frac{\pm 0.4}{20} \right) = 4 - (\pm 0.06)$$

where the variations are assumed to both have the same sign (correlated). You can see that this ratio error is 1.5% of the desired current ratio or gain.

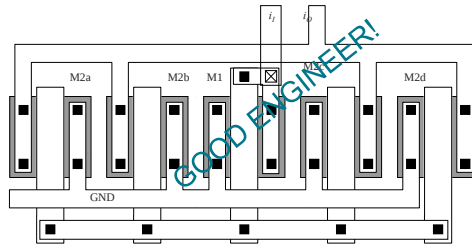
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Improved Layout of Current Mirrors

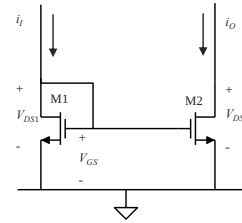
Using the following layout technique improves the ratio error of a current amplifier having the specifications of the previous example.

Reason: The actual widths of M1 and M2 are $W1 = 5 \pm 0.1 \mu\text{m}$ and $W2 = 4(5 \pm 0.1) \mu\text{m}$. The ratio of $W2$ to $W1$, and consequently the current gain, is given and is for all practical purposes independent of layout error.

$$\frac{i_O}{i_I} = \frac{4(5 \pm 0.1)}{5 \pm 0.1} = 4$$



Layout of 4:1 current mirror



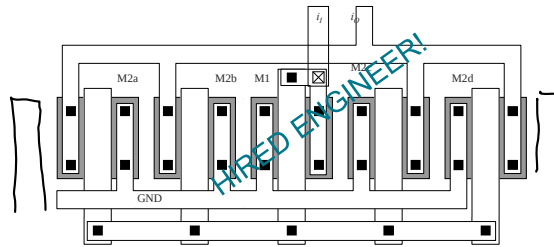
Current mirror

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Further Layout Improvements

Further layout improvements can be made:

- Add dummy devices to either side to reduce LOD effects.
- Add antenna diodes.
- Avoid proximity effects.
- Lay out using sufficient metal to avoid IR drops.



Layout of 4:1 current mirror

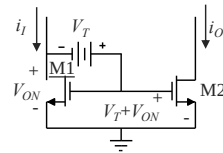
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MOS Current Mirror with Reduced $V_{\min}(\text{in})$

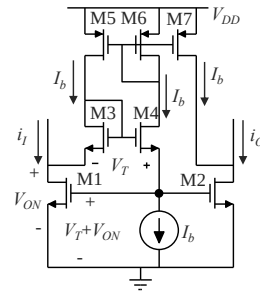
Minimum input voltage of the basic current mirror is $V_{\min}(\text{in}) = V_T + V_{ON}$ (Acceptable, but it can be reduced to V_{ON} . The basic idea is shown in the top schematic.)

Here are the ideal values:

- Minimum output voltage: $V_{\min}(\text{out}) = V_O$
- Output resistance: $R_{out} = \frac{1}{\lambda I_D}$
- Input resistance: $R_{in} = \frac{1}{g_m}$
- Current gain accuracy is poor because $V_{DS1} \neq V_{DS2}$.



Conceptual schematic for reducing $V_{\min}(\text{in})$



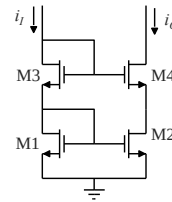
Actual schematic for reducing $V_{\min}(\text{in})$

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MOS Cascode Current Mirror

Improving the output resistance (at the expense of larger minimum output voltage)

- R_{out} : $v_{out} = r_{ds4}(i_{out} - g_{m4}v_{gs4}) + r_{ds2}(i_{out} - g_{m2}v_{gs2})$
But $i_{in} = 0$ so that
 $v_1 = v_3 = 0 \Rightarrow v_{gs4} = -v_{s4} = -i_{out}r_{ds2}$ and $v_{gs2} = 0$
 $\therefore v_{out} = i_{out}[r_{ds4} + r_{ds2} + g_{m4}r_{ds2}r_{ds4}] \approx r_{ds2}g_{m4}r_{ds4}$



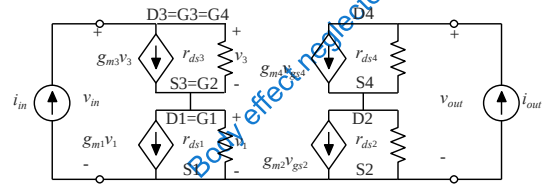
MOS cascode current mirror schematic

- R_{in} : $R_{in} = \frac{1}{g_{m3}} \parallel r_{ds3} + \frac{1}{g_{m1}} \parallel r_{ds1} \approx \frac{1}{g_{m1}} + \frac{1}{g_{m3}} \approx \frac{2}{g_m}$

- $V_{MIN}(out) = V_T + 2V_{ON}$

- $V_{MIN}(in) = 2(V_T + V_{ON})$

- Current gain match is excellent because $v_{ds1} = v_{ds2}$



Small-signal model of MOS cascode current mirror

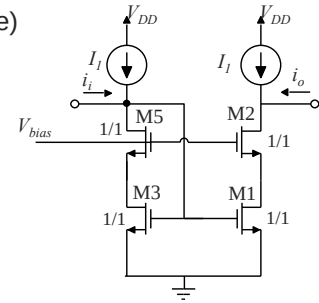
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Large Output Swing Cascode Current Mirror

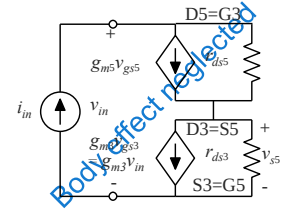
Improving the output resistance (with little impact on minimum output voltage)

- $R_{out} \approx g_{m2} r_{ds2} r_{ds1}$
- R_{in} : $R_{in} = ? v_{in} = r_{ds5}(i_{in} - g_{m5} v_{gs5}) + v_{s5} = r_{ds5}(i_{in} + g_{m5} v_{s5}) + v_{s5}$
 $= r_{ds5} i_{in} (1 + g_{m5} r_{ds5}) + v_{s5}$
 But $v_{s5} = r_{ds3}(i_{in} - g_{m3} v_{in})$
 $\therefore v_{in} = r_{ds5} i_{in} + (1 + g_{m5} r_{ds5}) r_{ds3} i_{in} - g_{m3} r_{ds3} (1 + g_{m5} r_{ds5}) v_{in}$

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{r_{ds5} + r_{ds3} + r_{ds3} g_{m5} r_{ds5}}{g_{m3} r_{ds3} (1 + g_{m5} r_{ds5})} \approx \frac{1}{g_{m3}}$$
- $V_{MIN}(out) = 2V_{ON}$
- $V_{MIN}(in) = V_T + V_{ON}$
- Current gain match is excellent because $v_{ds1} = v_{ds2}$.



Large output swing cascode current mirror schematic

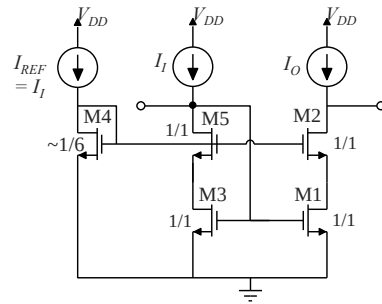


Small-signal model of large output swing cascode current mirror

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Biasing of Large Output Swing Cascode Current Mirror

- M4 is sized to keep M1, M3, and M5 out of triode region.
- With ideal square law devices and no body effect, current density in M4 should be four times that in M1.
- In practice, six times the current density is more appropriate.



Large output swing cascode current mirror schematic

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Self-Biased Cascode Current Mirror

Improving the output resistance:

- $R_{out} \approx g_{m4} r_{ds4} r_{ds2}$

- R_{in} :

$$v_{in} = i_{in} R + r_{ds3} (i_{in} - g_{m3} v_{gs3}) + r_{ds1} (i_{in} - g_{m1} v_{gs1})$$

But $v_{gs1} = v_{in} - i_{in} R$ and

$$v_{gs3} = v_{in} - r_{ds1} (i_{in} - g_{m1} v_{gs1}) = v_{in} - r_{ds1} i_{in} + g_{m1} r_{ds1} (v_{in} - i_{in} R)$$

$$\therefore v_{in} = i_{in} R + r_{ds3} - g_{m3} r_{ds3} [v_{in} - r_{ds1} i_{in} + g_{m1} r_{ds1} (v_{in} - i_{in} R)] + r_{ds1} [i_{in} - g_{m1} (v_{in} - i_{in} R)]$$

$$v_{in} [1 + g_{m3} r_{ds3} + g_{m1} r_{ds1} g_{m3} r_{ds3} + g_{m1} r_{ds1}]$$

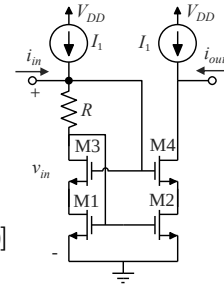
$$= i_{in} [R + r_{ds1} + r_{ds3} + g_{m3} r_{ds3} r_{ds1} + g_{m1} r_{ds1} g_{m3} r_{ds3} R]$$

$$R_{in} = \frac{R + r_{ds1} + r_{ds3} + g_{m3} r_{ds3} r_{ds1} + g_{m1} r_{ds1} g_{m3} r_{ds3} R}{1 + g_{m3} r_{ds3} + g_{m1} r_{ds1} g_{m3} r_{ds3} + g_{m1} r_{ds1}} \approx \frac{1}{g_{m1}} + R$$

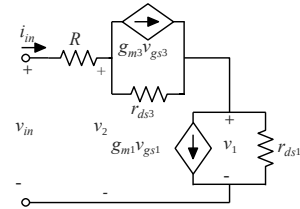
- $V_{MIN}(in) = V_T + 2V_{ON}$

- $V_{MIN}(out) = 2V_{ON}$

- Current gain matching is excellent, but resistor needs to be chosen carefully and can be large.



Self-biased cascode current mirror schematic



Small-signal model of self-biased cascode current mirror

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Wilson MOS Current Mirror

This current mirror uses negative series feedback to achieve higher output resistance.

R_{out} :

$$v_{out} = r_{ds2}(i_{out} - g_{m3}v_{gs3}) + v_{gs2}$$

$$v_{gs2} = \frac{i_{out}}{g_{m2} + g_{ds2}} = \frac{r_{ds2}i_{out}}{1 + g_{m2}r_{ds2}}$$

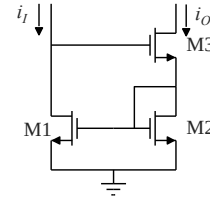
$$v_{gs3} = -g_{m1}r_{ds1}v_{gs2} - v_{gs2} = -(1 + g_{m1}r_{ds1})v_{gs2}$$

$$\therefore v_{out} = r_{ds2}i_{out} + g_{m3}r_{ds2}(1 + g_{m1}r_{ds1})v_{gs2}$$

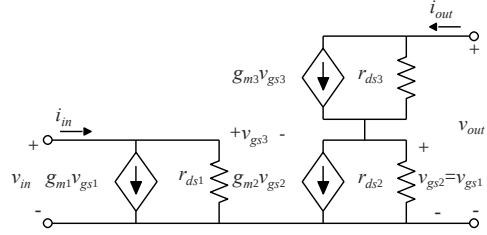
$$= i_{out} \left[r_{ds3} + r_{ds2} \left(\frac{1 + g_{m3}r_{ds2} + g_{m1}r_{ds1}g_{m3}r_{ds3}}{1 + g_{m2}r_{ds2}} \right) \right]$$

$$R_{out} = r_{ds3} + r_{ds2} \left(\frac{1 + g_{m3}r_{ds2} + g_{m1}r_{ds1}g_{m3}r_{ds3}}{1 + g_{m2}r_{ds2}} \right)$$

$$\approx \frac{g_{m1}r_{ds1}g_{m3}r_{ds3}}{g_{m2}}$$



Wilson current mirror schematic



Small-signal model of Wilson current mirror

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Wilson MOS Current Mirror (continued)

- R_{in} :
$$i_{in} = g_{m1}v_{gs1} = \frac{g_{m1}g_{m3}v_{gs2}}{g_{m2} + g_{ds2} + g_{ds3}} \approx \frac{g_{m1}g_{m3}v_{gs3}}{g_{m2}}$$

$$v_{gs3} = v_{in} - v_{gs1} = v_{in} - \frac{g_{m1}g_{m3}v_{gs3}}{g_{m2}} \Rightarrow v_{gs3} = \frac{v_{in}}{1 + \frac{g_{m1}g_{m3}}{g_{m2}}}$$

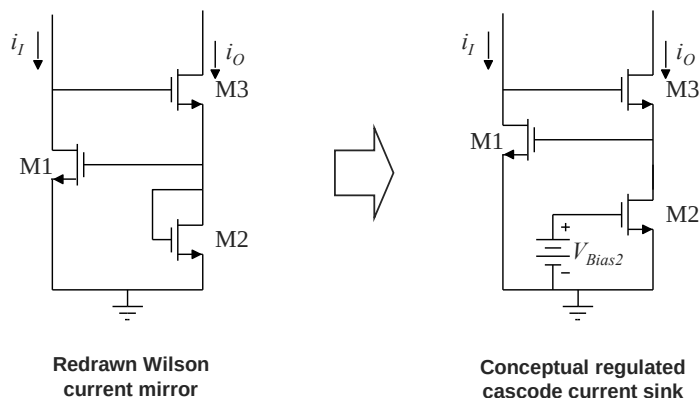
$$\therefore i_{in} \approx \frac{g_{m1}g_{m3}v_{in}}{g_{m2} + g_{m3}} \Rightarrow R_{in} = \frac{g_{m2} + g_{m3}}{g_{m1}g_{m3}}$$

- $V_{MIN}(in) = 2(V_T + V_{ON})$
- $V_{MIN}(out) = V_T + 2V_{ON}$
- Current gain matching is poor: $v_{DS1} \neq v_{DS2}$. Therefore, the enhanced version uses a fourth diode-connected transistor to balance V_{ds} .

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Regulated Cascode Current Mirror

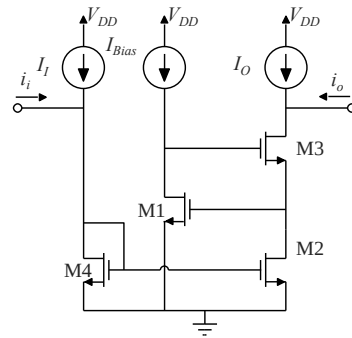
Evolution of regulated cascode current mirror from Wilson current mirror



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MOS Regulated Cascode Current Mirror

- $R_{out} \approx g_m^2 r_{ds}^3$
- $R_{in} \approx \frac{1}{g_{m4}}$
- $V_{MIN}(out) = V_T + 2V_{ON}$
- $V_{MIN}(in) = V_T + V_{ON}$
- Current gain matching is good as long as $v_{DS4} = v_{DS2}$.



Regulated cascode
current sink

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MOS Current Mirror Final Points

- Regulated cascodes are popular in opamps to achieve high voltage gain.
- Bandwidth of mirror is typically $f_t/(1+N)$, where f_t is the unity gain frequency of the input device and N is the mirror gain.



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MOS Current Mirror Summary

Current mirror type	Accuracy	Resistance input	Resistance minimum	Output voltage	Minimum input voltage
Simple	Poor	r_{ds}	$1/g_m$	V_{ON}	$V_T + V_{ON}$
Cascode	Excellent	$g_m r_{ds}^2$	$2/g_m$	$V_T + 2V_{ON}$	$2(V_T + V_{ON})$
Wide output swing cascode	Excellent	$g_m r_{ds}^2$	$1/g_m$	$2V_{ON}$	$V_T + V_{ON}$
Self-biased	Excellent	$g_m r_{ds}^2$	$R + 1/g_m$	$2V_{ON}$	$V_T + 2V_{ON}$
Wilson	Poor	$g_m r_{ds}^2$	$2/g_m$	$2(V_T + V_{ON})$	$V_T + 2V_{ON}$



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Summary: MOS Current Mirrors

Ideal current mirrors

- Output current is linearly related to the input current, $i_{out} = A_i i_{in}$.
- Input resistance is zero.
- Output resistance is infinity.
- Bandwidth is infinite.
- No added noise.

Influences on accuracy

- Channel modulation parameter (λ)
- Transistor mismatch
- Transistor aspect ratios

Types of current mirrors

- Basic MOS
- Cascode MOS
- Large output swing cascode MOS
- Self-biased cascode MOS
- Wilson MOS
- Regulated cascode MOS



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Introduction: Differential Amplifiers

- Large-signal analysis
 - Transconductance
 - Voltage-transfer characteristic
- Small-signal analysis
 - Differential-mode and common-mode analysis
 - Common-mode rejection ratio (CMRR)
 - Input common-mode range (ICMR)
- Architectures
 - Folding cascode
 - Linear transconductor
 - Cross-coupled differential amplifier



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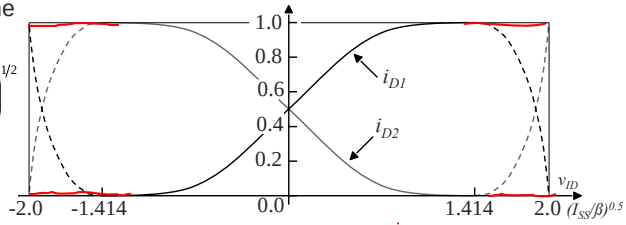
Large-Signal Transconductance (continued)

Defining equations (assuming that the MOSFETs are in saturation)

$$v_{ID} = v_{GS1} - v_{GS2} = \left(\frac{2i_{D1}}{\beta} \right)^{1/2} - \left(\frac{2i_{D2}}{\beta} \right)^{1/2}$$

and

$$\left(\beta = \frac{K_1 W_1}{L_1} \right) \quad I_{SS} = i_{D1} + i_{D2}$$



Plot of differential currents i_{D1} and i_{D2}

Solution

$$i_{D1} = \frac{I_{SS}}{2} + \frac{I_{SS}}{2} \left(\frac{\beta v_{ID}^2}{I_{SS}} - \frac{\beta^2 v_{ID}^4}{4I_{SS}^2} \right)^{1/2}$$

and

$$i_{D2} = \frac{I_{SS}}{2} - \frac{I_{SS}}{2} \left(\frac{\beta v_{ID}^2}{I_{SS}} - \frac{\beta^2 v_{ID}^4}{4I_{SS}^2} \right)^{1/2}$$

which are valid for $v_{ID} < (2I_{SS}/\beta)^{1/2}$

Differentiating i_{D1} (or i_{D2}) with respect to v_{ID} and setting $V_{ID} = 0V$ gives

$$g_m = \hat{i}_{D1} / \hat{v}_{ID}(v_{ID} = 0) = (\beta I_{SS} / 4)^{1/2} = \left(\frac{K_1' I_{SS} W_1}{4L_1} \right)^{1/2} \quad (\text{half the } g_m \text{ of an inverting amplifier})$$

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Large-Signal Voltage-Transfer Characteristics

MOS differential amplifier with resistor loads and MOSFETs in the saturation region:

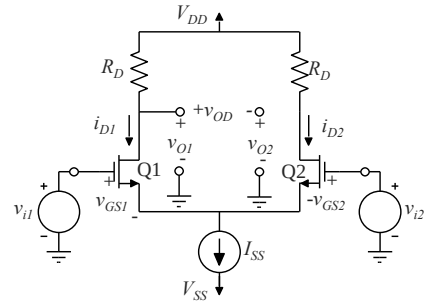
The differential output voltage is:

$$\begin{aligned} v_{OD} &= v_{DD} - i_{D1}R_D - v_{DD} + i_{D2}R_D \\ &= R_D(i_{D2} - i_{D1}) = R_D \left(I_{SS} \sqrt{\frac{\beta v_{ID}^2}{I_{SS}} - \frac{\beta^2 v_{ID}^4}{4I_{SS}^2}} \right) \end{aligned}$$

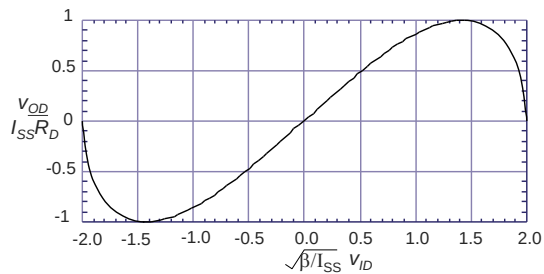
Notice that values of:

$$\sqrt{\frac{\beta}{I_{SS}}}$$

can range from 0.2-4V, which means that the linear range of MOS differential amplifiers is ± 0.2 -4V compared to ± 0.1 V for BJT differential amplifiers.



Differential amplifier schematic



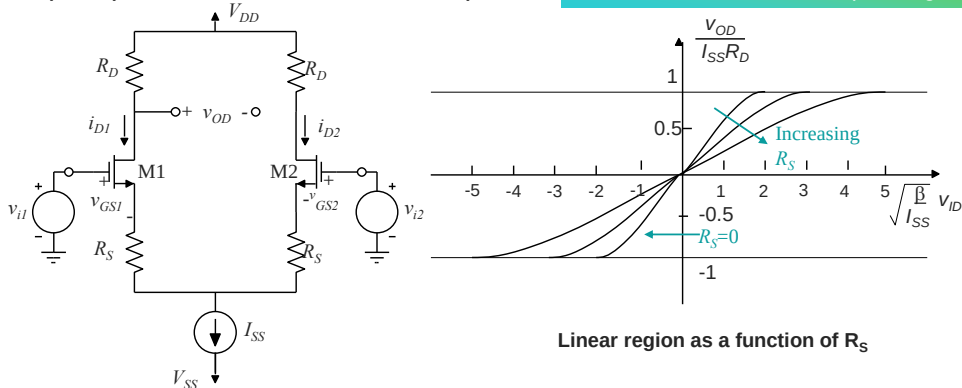
Plot of normalized differential output voltage

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Source Degeneration: MOS Differential Amplifier

Source degeneration increases the range over which the source-coupled pair behaves as a linear amplifier:

Note: Does not increase output range.



Linear region as a function of R_S

Differential amplifier with source degeneration schematic

Due to the increased linear range of MOSFET differential amplifiers, source degeneration is not as useful as for BJT differential amplifiers because the gain is lost in making the linearity-gain trade-off. (Gain is much more important to the MOSFET than to the BJT because of smaller values of g_m .)

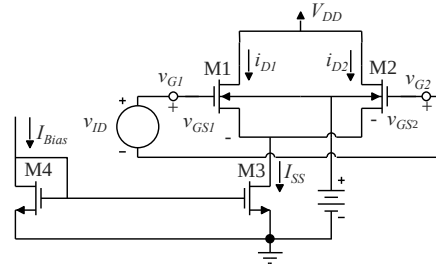
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Large-Signal Transconductance

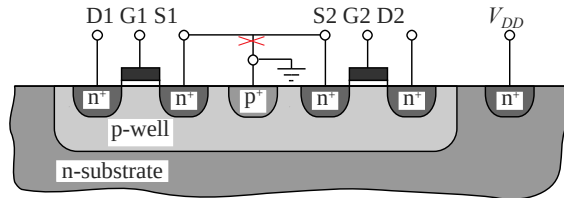
Consider the following n-channel differential amplifier (sometimes called a *source-coupled pair*). Where should bulk be connected? Consider a p-well CMOS technology:

- **Bulks connected to the well:** no modulation of V_T but large common-mode parasitic capacitance.
- **Bulks connected to ground:** smaller common-mode parasitic capacitors, but modulation of V_T .

If the technology is n-well CMOS, the bulks must be connected to ground.



Source-coupled pair schematic

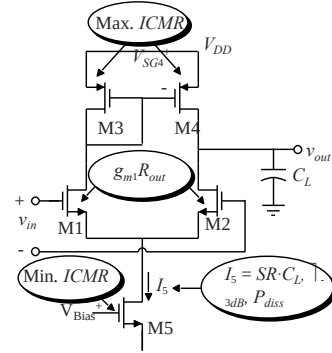


Source-coupled pair cross section

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Differential Amplifier: Design Process

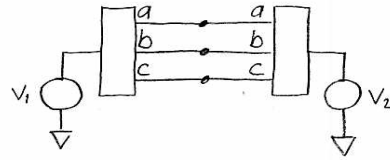
1. Pick I_{SS} to satisfy the slew rate, knowing C_L or the power dissipation.
2. Check to see if R_{out} satisfies the frequency response. If not change I_{SS} or modify circuit.
3. Design W_3/L_3 (W_4/L_4) to satisfy the upper $ICMR$.
4. Design W_1/L_1 (W_2/L_2) to satisfy the gain.
5. Design W_5/L_5 to satisfy the lower $ICMR$.
6. Iterate where necessary.



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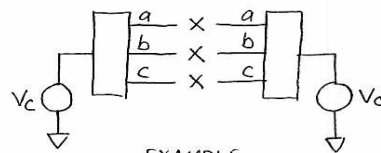
Half-Circuit Concept: Symmetry Analysis

Consider two completely symmetrical circuits;
a,b,c are connected
points of symmetry

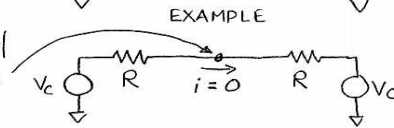


COMMON MODE

If $V_1 = V_2 = V_c$ (both circuits have the same input;
symmetric excitation) $\Rightarrow$ then we can open all
leads between points of
symmetry without affecting
circuit operation; no
current flows across
any connection



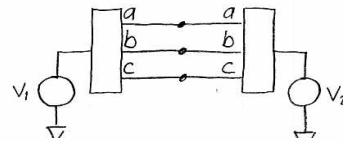
EXAMPLE: $i=0$ (no voltage drop
across R_s due to symmetrical
excitation. We can open this
connection without
affecting performance



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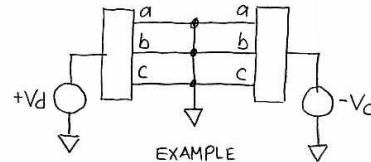
Symmetry Analysis (continued)

Consider two completely symmetrical circuits;
a,b,c are connected
points of symmetry

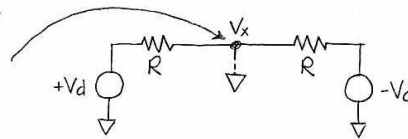


DIFFERENTIAL MODE

If $V_1 = -V_2$ (antisymmetric excitation) $\Rightarrow$ then we can
signal ground all leads
between points of
symmetry



EXAMPLE: by voltage divider,
 $V_x = 0$ regardless of V_d .
Therefore we can signal
ground this connection
without affecting
circuit performance

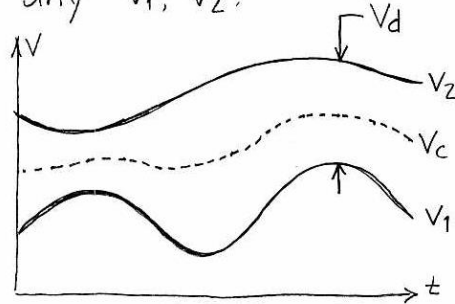


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Half-Circuit Concept: Symmetry Analysis

Any two V or I signals can be considered as the sum of common mode +/- differential signal.

Consider any V_1, V_2 :



Define

$$V_c = \frac{V_1 + V_2}{2}$$

$$V_d = V_2 - V_1$$

Straightforward to verify that

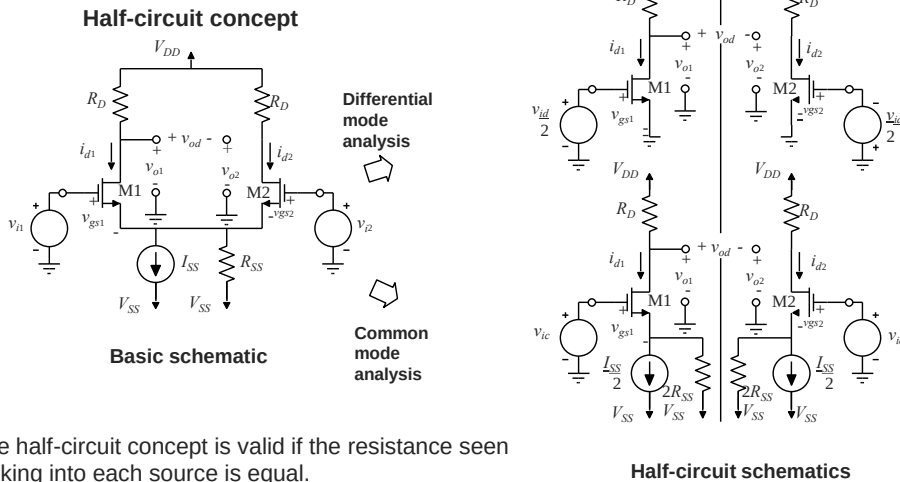
$$V_1 = V_c - \frac{V_d}{2}$$

$$V_2 = V_c + \frac{V_d}{2}$$

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Differential-Mode and Common-Mode Analysis

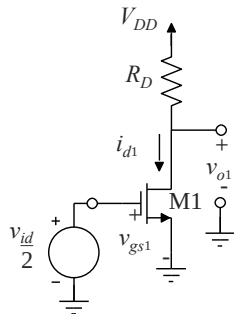
The small-signal performance of a differential amplifier can be separated into differential-mode and common-mode analysis, enabling you to take advantage of the following simplifications:



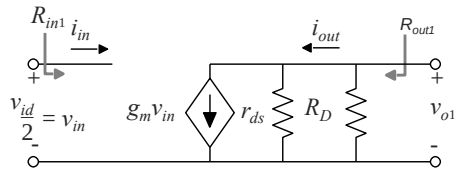
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Small-Signal Differential-Mode Performance



Differential half-circuit schematic



Differential half-circuit model

Half-circuit performance

$$R_{in1} = \infty, \quad R_{out1} \approx R_D \quad \text{and} \quad \frac{v_{o1}}{v_{id}} = \frac{-g_{m1}R_D}{2}$$

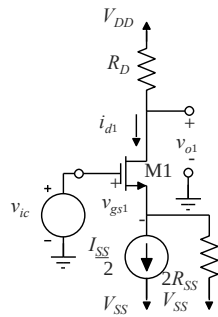
Differential performance

$$R_{id} = \infty, \quad R_{od} = R_{out1} + R_{out2} \quad \text{and} \quad \frac{v_{od}}{v_{id}} = \frac{v_{o1}}{v_{id}} - \frac{v_{o2}}{v_{id}} = \frac{-g_{m1}R_D}{2} - \frac{+g_{m2}R_D}{2} = -g_{m1}R_D$$

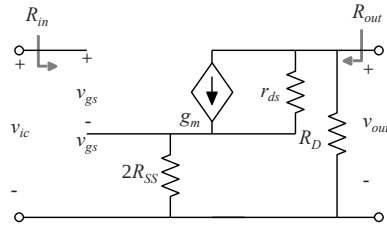
where $g_{m1} = g_{m2}$.

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Small-Signal Common-Mode Performance



Common half-circuit schematic



Common half-circuit model

Half-circuit performance

$$R_{in1} = \infty, \quad R_{out} = R_D \parallel (r_{ds} [1 + (g_m + g_{mbs}) 2R_{SS}] + 2R_{SS}) \quad \text{and} \quad \frac{v_{o1}}{v_{ic}} = -\frac{g_m R_D}{1 + g_m 2R_{SS}}$$

Common-mode performance

$$R_{ic} = \infty, \quad R_{out} = R_D \parallel (r_{ds} [1 + (g_m + g_{mbs}) 2R_{SS}] + 2R_{SS}) \quad \text{and} \quad \frac{v_{oc}}{v_{ic}} = -\frac{g_m R_D}{1 + g_m 2R_{SS}}$$

where $g_{m1} = g_{m2}$ and $r_{ds1} = r_{ds2}$

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Common-Mode Rejection Ratio

The common-mode rejection ratio (CMRR) is a measure of the differential amplifier's ability to reject the common-mode signal and amplify the differential-mode signal.

$$CMRR = \left| \frac{A_{dm}}{A_{cm}} \right| = \left| \frac{v_{o1}/v_{id}}{v_{o1}/v_{ic}} \right| = \frac{\frac{g_{m1}R_D}{2}}{\frac{g_{m1}R_D}{1 + g_{m1}2R_{SS}}} \approx g_{m1}R_{SS}$$

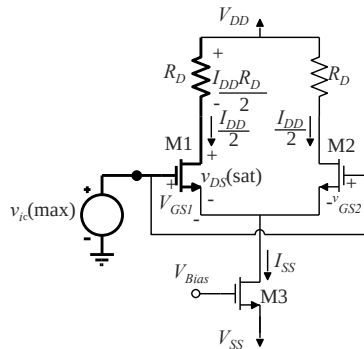
Thus, the larger the input transconductance, or R_{SS} , the larger the common-mode rejection ratio.



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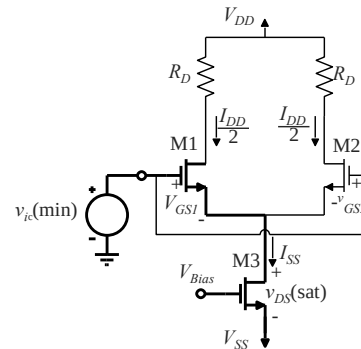
Input Common-Mode Voltage Range

The input common-mode voltage range (ICMR) is the range of common mode input voltages over which the differential amplifier amplifies the differential signal without significant change. Consider the following:



Maximum input common-mode voltage:

$$\begin{aligned} v_{ic}(\max) &= V_{DD} - 0.5I_{SS}R_D - v_{DS1}(\text{sat}) + v_{GS1} \\ &= V_{DD} - 0.5I_{SS}R_D + V_{T1} \end{aligned}$$



Minimum input common-mode voltage:

$$v_{ic}(\min) = V_{SS} + v_{DS3}(\text{sat}) + v_{GS1}$$

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Input Common-Mode Voltage Range (continued)

The ICMR is found by setting $v_{ID} = 0$ and varying v_{IC} until one of the transistors leaves saturation.

Highest common-mode voltage:

Path from $G1$ through $M1$ and $M3$ to V_{DD} is

$$V_{IC}(\max) = V_{G1}(\max) = V_{G2}(\max) = V_{DD} - V_{SG3} - V_{DS1}(\text{sat}) + V_{GS1}$$

or

$$V_{IC}(\max) = V_{DD} - V_{SG3} + V_{GS1}$$

Path from $G2$ through $M2$ and $M4$ to V_{DD} is

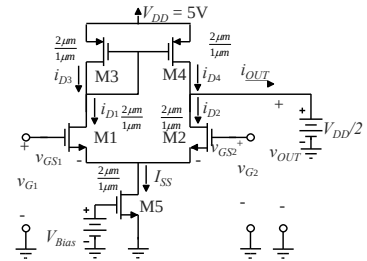
$$V_{IC}(\max) = V_{DD} - V_{SD4}(\text{sat}) - V_{SD2}(\text{sat}) + V_{GS2} = V_{DD} - V_{SD4}(\text{sat}) + V_{TN}$$

$$\therefore V_{IC}(\max) = V_{DD} - V_{SG3} + V_{GS1}$$

Lowest common-mode voltage (assume a V_{SS} for generality):

$$V_{IC}(\min) = V_{SS} + V_{DS5}(\text{sat}) + V_{GS1} = V_{SS} + V_{DS5}(\text{sat}) + V_{GS2}$$

where it has been assumed that $V_{GS1} = V_{GS2}$ during changes in the input common mode voltage.



Differential amplifier
with active load

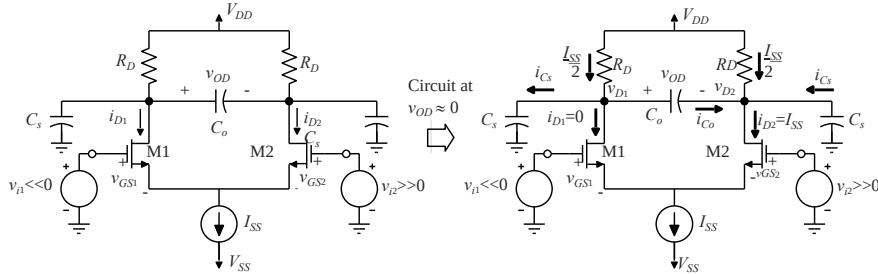
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Slew Rate

Because the current that is available to charge a capacitor is constant, the slew rate (SR) is a voltage rate of change:

$$\text{Slew rate} = \text{SR} = \frac{dV_C}{dt} = \frac{i_C}{C}$$

where i_C and v_C are the current through and voltage across a capacitor C . A differential amplifier that has a capacitive load experiences the slew rate as follows:



Notice that the current in M1 is 0 and M2 is I_{SS} . The slew rate can be found from

$$0.5I_{SS} = C_s \frac{dv_{D1}}{dt} + C_o \frac{d(v_{D1} - v_{D2})}{dt} \quad \text{and} \quad 0.5I_{SS} = -C_s \frac{dv_{D2}}{dt} + C_o \frac{d(v_{D1} - v_{D2})}{dt}$$

Adding these two equations gives

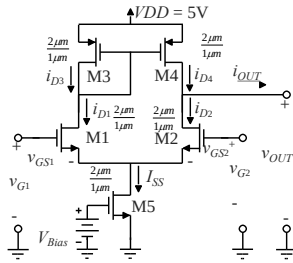
$$I_{SS} = 2C_o \frac{d(v_{D1} - v_{D2})}{dt} + C_s \frac{d(v_{D1} - v_{D2})}{dt} = (2C_o + C_s) \frac{d(v_{D1} - v_{D2})}{dt} \Rightarrow \text{SR} = \frac{I_{SS}}{2C_o + C_s} = \frac{0.5I_{SS}}{C_o + 0.5C_s}$$



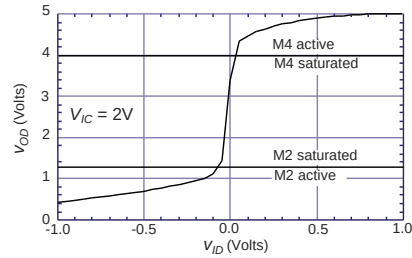
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Voltage-Transfer Characteristic

The transconductance characteristics of the MOSFET differential amplifier that do *not* depend upon the load have been described. Now consider the following load:



Differential amplifier with active load



Transfer function

M2 is in saturation when:

$$v_{DS2} \geq v_{GS2} - V_{TN} \rightarrow v_{OUT} - v_{S1} \geq V_{IC} - v_{S1} - V_{TN} \rightarrow v_{OUT} \geq V_{IC} - V_{TN}$$

where the region of transition for M2 is assumed to be close to $v_{ID} = 0V$.

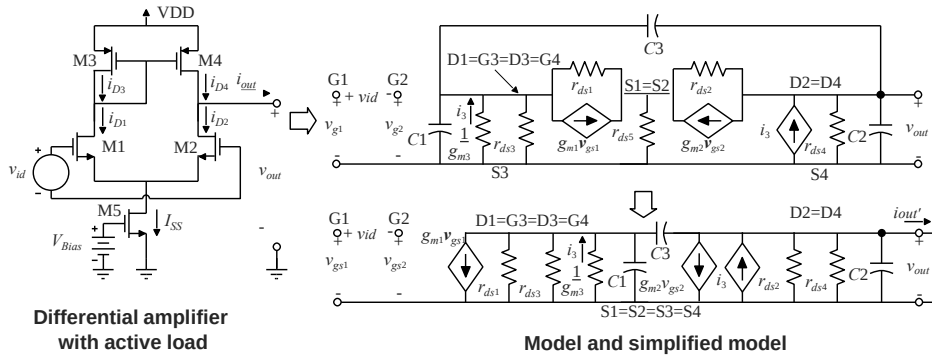
M4 is in saturation when:

$$v_{SD4} \geq v_{SG4} - |V_{TP}| \rightarrow V_{DD} - v_{OUT} \geq v_{SG4} - |V_{TP}| \rightarrow v_{OUT} \leq V_{DD} - v_{SG4} + |V_{TP}|$$

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Small-Signal Analysis: Differential Mode

A requirement for differential-mode operation is that the differential amplifier is balanced:



Differential transconductance: Assume that the output of the differential amplifier is an AC short:

$$i'_{out} = \frac{g_{m1}g_{m3}r_{p1}}{1 + g_{m3}r_{p1}} v_{gs1} - g_{m2}v_{gs2} \approx g_{m1}v_{gs1} - g_{m2}v_{gs2} = g_{md}v_{id}$$

where $g_{m1} = g_{m2} = g_{md}$, $r_{p1} = r_{ds1} || r_{ds3}$, and i'_{out} designates the output current into a short circuit.

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Small-Signal Analysis: Differential Mode (continued)

Output resistance

$$r_{out} = \frac{1}{g_{ds2} + g_{ds4}} = r_{ds2} \parallel r_{ds4}$$

Differential voltage gain

$$A_v = \frac{v_{out}}{v_{id}} = \frac{g_{md}}{g_{ds2} + g_{ds4}}$$

Assuming that all transistors are in saturation and replace the small-signal parameters of g_m and r_{ds} in terms of their large-signal model equivalents, the voltage gain can be written as

$$A_v = \frac{v_{out}}{v_{id}} = \frac{(K'_1 I_{SS} W_1 / L_1)^{1/2}}{(\lambda_2 + \lambda_4)(I_{SS}/2)} = \frac{2}{\lambda_2 + \lambda_4} \left(\frac{K'_1 W_1}{I_{SS} L_1} \right)^{1/2} \propto \frac{1}{\sqrt{I_{SS}}}$$

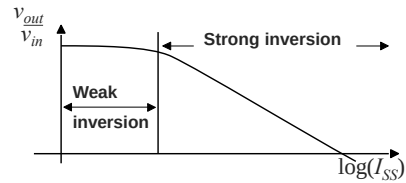
Note: The small-signal gain is inversely proportional to the square root of the bias current!

If $W_1/L_1 = 2\mu\text{m}/1\mu\text{m}$ and $I_{SS} = 50\mu\text{A}$ ($10\mu\text{A}$)

$$A_v(\text{n-channel}) = 46.6\text{V/V} (104.23\text{V/V})$$

$$A_v(\text{p-channel}) = 31.4\text{V/V} (70.27\text{V/V})$$

$$r_{out} = \frac{1}{g_{ds2} + g_{ds4}} = \frac{1}{25\mu\text{A} \cdot 0.9\text{V}^{-1}} = 0.444\text{M}\Omega (0.222\text{M}\Omega)$$

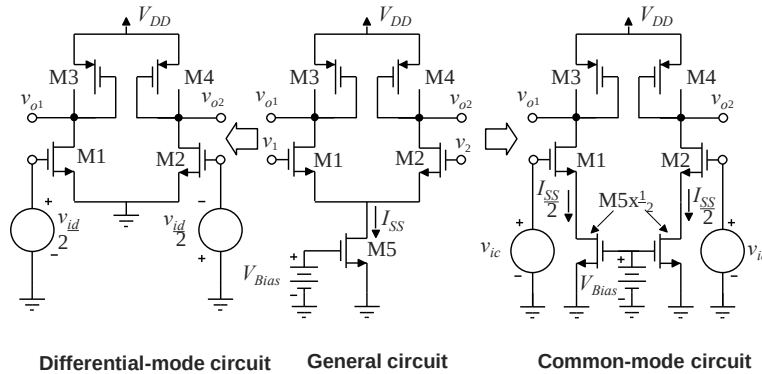


Voltage gain as a function of bias current

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Small-Signal Analysis

The common-mode gain of the differential amplifier with a current mirror load is ideally zero. To illustrate the common-mode gain, a different type of load is needed. Consider the following:



Differential-mode analysis is

$$\frac{v_{o1}}{v_{id}} \approx -\frac{g_{m1}}{2g_{m3}} \quad \text{and} \quad \frac{v_{o2}}{v_{id}} \approx +\frac{g_{m2}}{2g_{m4}}$$

Notice that these voltage gains are half of the active-load inverter voltage gain.

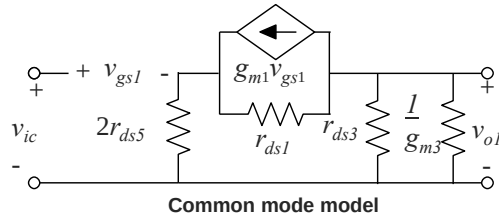
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Small-Signal Analysis (continued)

Common-mode analysis: Assume that r_{ds1} is large and can be ignored, which greatly simplifies the analysis.

$$\therefore v_{gs1} = v_{g1} - v_{s1} = v_{ic} - 2g_{m1}r_{ds5}v_{gs1}$$

Solving for v_{gs1} gives
$$v_{gs1} = \frac{v_{ic}}{1 + 2g_{m1}r_{ds5}}$$



The single-ended output voltage, v_{o1} , as a function of v_{ic} can be written as

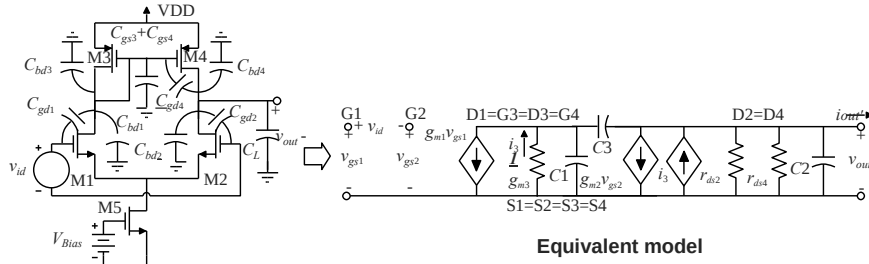
$$\frac{v_{o1}}{v_{ic}} = -\frac{g_{m1}[r_{ds3} \parallel 1/g_{m3}]}{1 + 2g_{m1}r_{ds5}} \approx -\frac{(g_{m1}/g_{m3})}{1 + 2g_{m1}r_{ds5}} \approx -\frac{g_{ds5}}{2g_{m3}}$$

The CMRR (common-mode rejection ratio) is

$$CMRR = \frac{|v_{o1}/v_{id}|}{|v_{o1}/v_{ic}|} = \frac{g_{m1}/2g_{m3}}{g_{ds5}/2g_{m3}} = g_{m1}r_{ds5}$$

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Frequency Response: Active-Load Diff Amp



Differential amplifier with parasitic capacitances

Ignore the zeros that occur due to C_{gd1} , C_{gd2} and C_{gd4} :

$$C_1 = C_{gd1} + C_{bd1} + C_{bd3} + C_{gs3} + C_{gs4} \quad C_2 = C_{bd2} + C_{bd4} + C_{gd2} + C_L \quad \text{and} \quad C_3 = C_{gd4}$$

If $C_3 \approx 0$, then you can write

$$v_{out}(s) \approx \frac{g_{m1}}{g_{ds2} + g_{ds4}} \left[\left(\frac{g_{m3}}{g_{m3} + sC_1} \right) V_{gs1}(s) - V_{gs2}(s) \right] \left(\frac{\omega_2}{s + \omega_2} \right) \quad \text{where} \quad \omega_2 = \frac{g_{gs2} + g_{ds4}}{C_2}$$

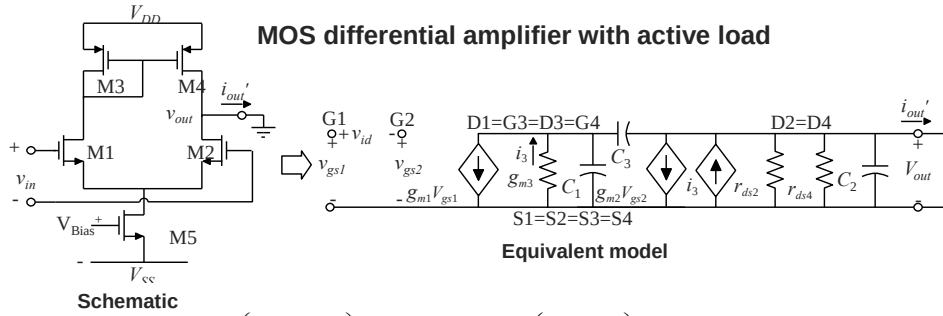
If you further assume that $g_{m3}/C_1 \gg (g_{ds2} + g_{ds4})/C_2 = \omega_2$

then the frequency response of the differential amplifier reduces to

$$\frac{V_{out}(s)}{V_{id}(s)} \cong \left(\frac{g_{m1}}{g_{ds2} + g_{ds4}} \right) \left(\frac{\omega_2}{s + \omega_2} \right)$$

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Frequency Response (continued)



$$i_{out}' = i_3 - g_{m2}v_{gs2} = - \left(\frac{\frac{1}{sC_1}}{\frac{1}{g_{m3}} + \frac{1}{sC_1}} \right) g_{m1}v_{gs1} - g_{m2}v_{gs2} = - \left(\frac{1}{s \frac{C_1}{g_{m3}} + 1} \right) g_{m1} \frac{v_{in}}{2} - g_{m2} \frac{v_{in}}{2}$$

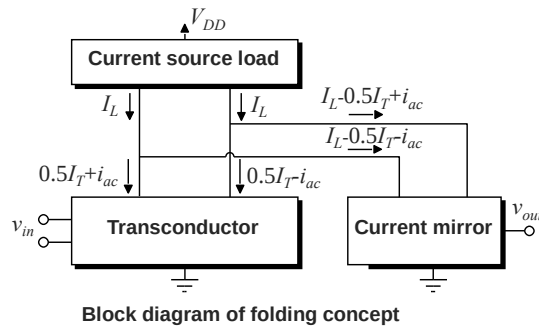
$$i_{out}' = -g_{m1} \left(\frac{s \frac{C_1}{g_{m3}} + 2}{s \frac{C_1}{g_{m3}} + 1} \right) \frac{v_{in}}{2} = -g_{m1} \left(\frac{s \frac{C_1}{g_{m3}} + 2}{s \frac{C_1}{g_{m3}} + 1} \right) \frac{v_{in}}{2} \text{ where } g_{m1} = g_{m2} = g_{m1}$$

The circuit has a zero at $-2g_{m3}/C_1$ and a pole at $-g_{m3}/C_1$. (These roots are not dominant. C_2 causes the -3 dB frequency.)



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Folding



Advantages

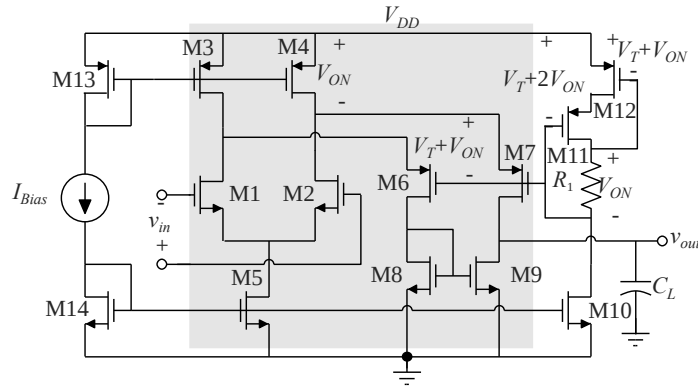
- Increases the upper common-mode voltage range
- Allows balanced loads
- Smaller power supplies

Disadvantages

- Introduces more poles (generally not dominant)
- I_L must be greater than $I_T + i_{ac}$ so the current mirror will not be starved
- More offsets, more noise, larger area, higher current consumption

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Folded MOS Differential Amplifiers



Folded differential amplifier schematic

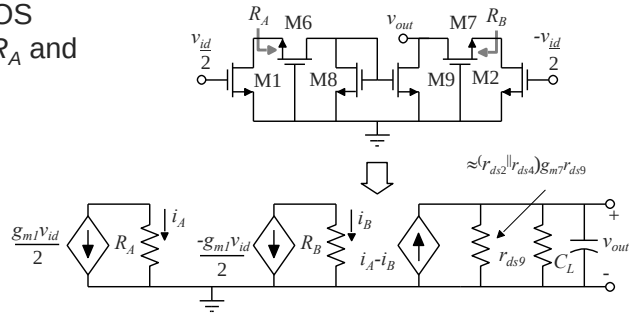
- Maximum input common-mode range is $V_{DD} - V_{ON} + V_T > V_{DD}$.
- Voltages at the drains of M1 and M2 are identical, which reduces the systematic offset.
- Impedance seen looking into the sources of M1 and M2 are nearly identical, thus the circuit is balanced.
- Only the high-impedance point is at the output that creates a dominant pole with C_L .

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Small-Signal Analysis of Folded Cascode

Recall from the cascode MOS amplifier that the resistors R_A and R_B can be expressed as:

$$R_A \approx \frac{1}{g_{m6}} \quad \text{and} \quad R_B \approx \frac{2}{g_{m7}}$$



Equivalent circuit and model of folded differential amplifier

$$\therefore v_{out} = \left(\frac{g_{m1} + g_{m2}}{2} \right) r_{ds9} \parallel [g_{m7} r_{ds7} (r_{ds2} \parallel r_{ds4})] v_{id} \approx g_{m1} r_{ds9} v_{id} \Rightarrow \frac{v_{out}}{v_{id}} = g_{m1} r_{ds9}$$

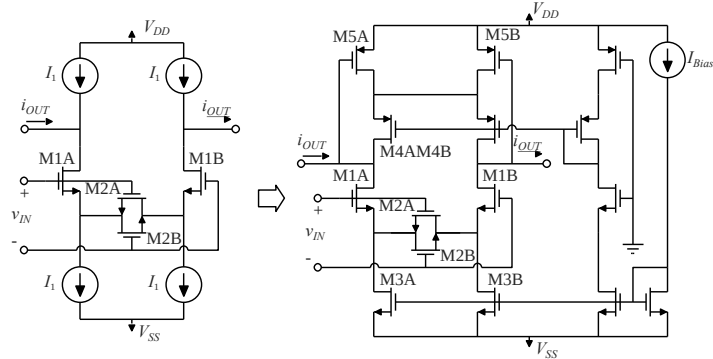
$$R_{out} \approx r_{ds9}$$

$$\text{-3 dB frequency is } \omega_{-3dB} = \frac{1}{r_{ds9} C_L}$$

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Linear CMOS Transconductors

In many applications, it is necessary to linearize the transconductance of a differential amplifier. This circuit* shows one method of doing this.



Simplified and complete schematic of linear transconductor

M2A and M2B are normally in the triode region.

∴ The small signal drain-source resistance of M2A and M2B is

$$r_{ds2} = r_{ds2A} = r_{ds2B} = \frac{1}{\beta_2(V_{GS1} - V_{TN})}$$

*F. Krummenacher and N. Joehl, "A 4-MHz CMOS Continuous-Time Filter with On-Chip Automatic Tuning," *IEEE Journal of Solid-State Circuits*, vol. 23, no. 3, June 1988, pp. 750-758.

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Linear CMOS Transconductors (continued)

The small signal transconductances of M1A and M1B are

$$g_{m1} = g_{m1A} = g_{m1B} = \beta_1(V_{GS1} - V_{TN})$$

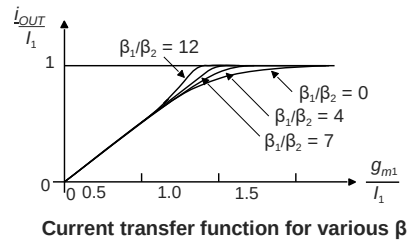
Using the half-circuit approach for differential operation, you can write that the small-signal output current is

$$i_{out} = g_{m1}v_{gs1} = g_{m1}\left(\frac{v_{in}}{2} - 0.5r_{ds2}i_{out}\right) \rightarrow i_{out} = \frac{0.5g_{m1}v_{in}}{1 + 0.5g_{m1}r_{ds2}} \rightarrow g(\text{eff}) = \frac{0.5g_{m1}}{1 + 0.5g_{m1}r_{ds2}}$$

Substituting the value of g_{m1} and r_{ds2} gives

$$g(\text{eff}) = \frac{0.5\beta_1(V_{GS1} - V_{TN})}{1 + \frac{0.5\beta_1(V_{GS1} - V_{TN})}{0.5\beta_2(V_{GS2} - V_{TN})}} = \frac{\beta_1\beta_2(V_{GS1} - V_{TN})}{\beta_1 + 2\beta_2} = \frac{\beta_1\beta_2}{\beta_1 + 2\beta_2} \sqrt{\frac{2I_1}{\beta_1}}$$

Select the ratio of β_1/β_2 to linearize the transconductance:



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Cross-Coupled Differential Amplifiers

This input stage is good for class B operation:

The voltage loop is

$$\begin{aligned} v_{i1} = v_{i2} &= -V_{GS1} + v_{GS1} + v_{SG4} - V_{SG4} \\ &= V_{SG3} - v_{SG3} - v_{GS2} + V_{GS2} \end{aligned}$$

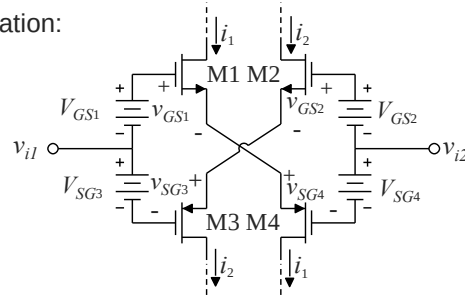
Using the notation for AC, DC, and total variables gives

$$v_{i2} - v_{i1} = v_{id} = (v_{sg1} + v_{gs4}) = -(v_{sg3} + v_{gs2})$$

If $M1 = M2 = M3 = M4$, then half of the differential input is applied across each transistor with the correct polarity:

$$\therefore i_1 = \frac{g_{m1}v_{id}}{2} = \frac{g_{m4}v_{id}}{2} \text{ and } i_2 = -\frac{g_{m2}v_{id}}{2} = -\frac{g_{m3}v_{id}}{2}$$

Note: For low-voltage designs, this input stage is often not well suited due to limited ICMR.

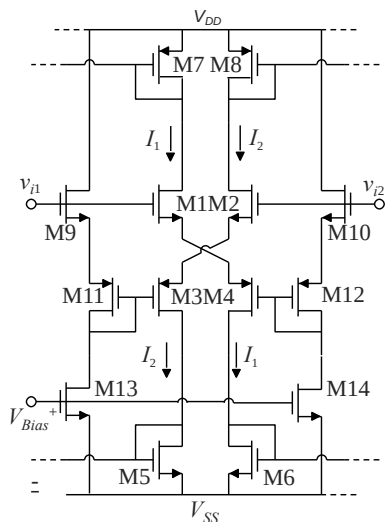


Simplified cross-coupled differential amplifier schematic

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Cross-Coupled Differential Amplifiers (continued)

Gain is halved because the input signal is dropped across two gate sources.



Cross-coupled differential amplifier schematic

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Summary: Differential Amplifiers

- Large-signal analysis
 - Transconductance
 - Voltage-transfer characteristic
- Small-signal analysis
 - Differential-mode and common-mode analysis
 - Common mode rejection ratio (CMRR)
 - Input common mode range (ICMR)
- Architectures
 - Folding cascode
 - Linear transconductor
 - Cross-coupled differential amplifier



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Introduction to Multistage Amplifiers

- Cascodes
- Cascades
- Folded cascodes
- Output stages
 - Source follower
 - Push-pull source follower
 - Push-pull common-source
 - Negative shunt feedback



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Cascodes

Why are cascode gain stages important?

- Higher open-loop gain
- Higher-output impedance
- Lower-input capacitance

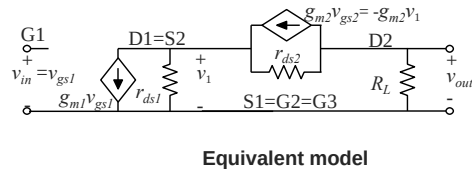
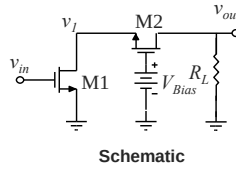


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MOS Cascode Amplifier

Small-signal performance
(assuming a load
resistance in the
drain of R_L):

$$R_{in} = \infty$$



Using nodal analysis, you can write

$$[g_{ds1} + g_{ds2} + g_{m2}]v_1 - g_{ds2}v_{out} = -g_{m1}v_{in} \quad \text{and} \quad -[g_{ds2} + g_{m2}]v_1 + (g_{ds2} + G_L)v_{out} = 0$$

Solving for v_{out}/v_{in} yields:
$$\frac{v_{out}}{v_{in}} = \frac{-g_{m1}(g_{ds2} + g_{m2})}{g_{ds1}g_{ds2} + g_{ds1}G_L + g_{ds2}G_L + G_Lg_{m2}} \cong \frac{-g_{m1}}{G_L} = -g_{m1}R_L$$

$$\frac{v_1}{v_{in}} = -g_{m2} \left[r_{ds2} \left\| \left(\frac{r_{ds2} + R_L}{1 + g_{m2}r_{ds2}} \right) \right] \approx -\frac{r_{ds2} + R_L}{r_{ds2}} = -\left(1 + \frac{R_L}{r_{ds2}} \right)$$

(R_L must be much less than r_{ds2} for the gain to be -1)

The small-signal output resistance is $r_{out} = [r_{ds1} + r_{ds2} + g_{m2}r_{ds1}r_{ds2}]R_L \cong R_L$

Note: Voltage gain can be greater than unity because of the positive feedback through the output resistance of M2.

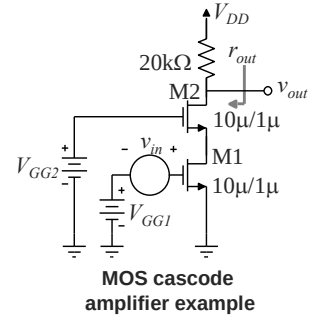


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MOS Cascode Example

A MOS cascode amplifier is shown. Assume all transistors are saturated and the $k_n' = 110 \mu\text{A}/\text{V}^2$, $V_T = 0.7\text{V}$, and $I_N = 0.04\text{V}^{-1}$. The value of V_{GG1} gives a DC bias current in both transistors of $100 \mu\text{A}$:

- Find the value of V_{GG2} that gives a minimum DC output voltage. What is the value of the minimum DC output voltage?
- Find the small-signal output resistance r_{out} .
- Find the small-signal voltage gain v_{out}/v_{in} .



Solution

$$\text{a. } V_{DS}(\text{sat}) = V_{ON} = \sqrt{\frac{2I_D}{K'(W/L)}} = \sqrt{\frac{2 \cdot 100 \mu\text{A}}{110 \mu\text{A}/\text{V}^2 \cdot 10}} = 0.4264\text{V}$$

$$\therefore V_{GS} = V_{ON} + V_T = 0.4264 + 0.7 = 1.1264\text{V} \Rightarrow V_{GG2} = 2V_{ON} + V_T = 1.553\text{V} \quad \text{also} \quad V_{MIN} = 2V_{ON} = 2 \cdot 0.4264 = 0.8528\text{V}$$

$$\text{b. } R_{out} = 20\text{k}\Omega \parallel g_{m2} r_{ds} \quad g_m = \sqrt{2I_D \cdot k' \frac{W}{L}} = \sqrt{2 \cdot 100 \cdot 110 \cdot 10} = 469 \mu\text{S}$$

$$r_{ds} = \frac{1}{100 \mu\text{A} \cdot 0.04} = 0.25\text{M}\Omega \quad \therefore R_{out} = 20\text{k}\Omega \parallel 29.3\text{M}\Omega \approx 20\text{k}\Omega$$

$$\text{c. The small signal voltage gain can be expressed as } \frac{v_{out}}{v_{in}} = -g_m R_{out} = -(469 \mu\text{S})(0.02\text{M}\Omega) = -9.38\text{V/V}$$

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MOS Cascode Amplifier Frequency Response

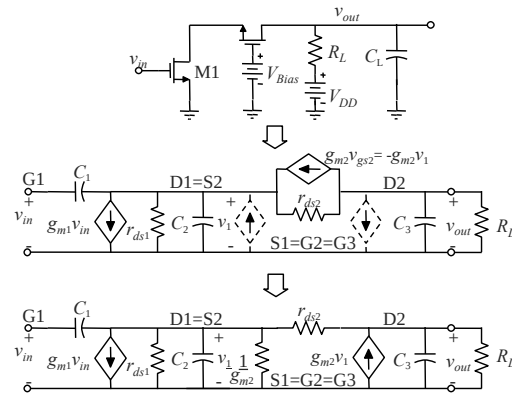
- Preparing the small-signal model:
- Small-signal model ($g_{m2}v_1$ has been rearranged and the substitution theorem applied) where:

$$C_1 = C_{gd1},$$

$$C_2 = C_{bd1} + C_{bs2} + C_{gs2}$$

and

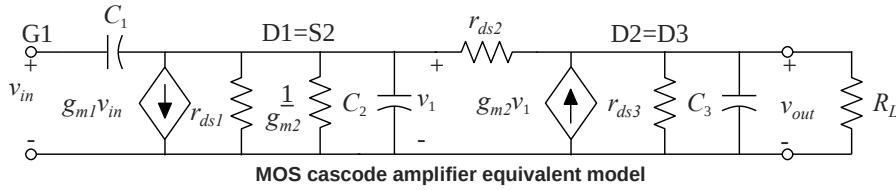
$$C_3 = C_{bd2} + C_{gd2} + C_L$$



MOS cascode amplifier schematic and equivalent model

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Frequency Response (continued)



The nodal equations now become

$$(g_{m2} + g_{ds1} + g_{ds2} + sC_1 + sC_2)v_1 - g_{ds2}v_{out} = -(g_{m1} - sC_1)v_{in}$$

and $-(g_{ds2} + g_{m2})v_1 + (g_{ds2} + G_L + sC_3)v_{out} = 0$

Solving for $v_{out}(s)/v_{in}(s)$ gives
$$\frac{v_{out}(s)}{v_{in}(s)} = \left(\frac{1}{1 + as + bs^2} \right) \left(\frac{-(g_{m1} - sC_1)(g_{ds2} + g_{m2})}{g_{ds1}g_{ds2} + G_L(g_{m2} + g_{ds1} + g_{ds2})} \right)$$

Where
$$a = \frac{C_3(g_{ds1} + g_{ds2} + g_{m2}) + C_2(g_{ds2} + G_L) + C_1g_{ds2}}{g_{ds1}g_{ds2} + G_L(g_{m2} + g_{ds1} + g_{ds2})}$$

and
$$b = \frac{C_3(C_1 + C_2)}{g_{ds1}g_{ds2} + G_L(g_{m2} + g_{ds1} + g_{ds2})}$$



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Simplified Method: Algebraic Expression for Two Poles

Assume that a general second-order polynomial can be written as

$$P(s) = 1 + as + bs^2 = \left(1 - \frac{s}{p_1}\right) \left(1 - \frac{s}{p_2}\right) = 1 - s \left(\frac{s}{p_1} + \frac{s}{p_2}\right) + \frac{s^2}{p_1 p_2}$$

Now if $|p_2| \gg |p_1|$, then $P(s)$ can be simplified as $P(s) \approx 1 - \frac{s}{p_1} + \frac{s^2}{p_1 p_2}$

Therefore, you can write p_1 and p_2 in terms of a and b as

$$p_1 = \frac{-1}{a} \quad \text{and} \quad p_2 = \frac{-a}{b}$$

Applying this to the previous problem gives $p_1 = \frac{-[g_{ds1}g_{ds2} + G_L(g_{m2} + g_{ds1} + g_{ds2})]}{C_3(g_{ds1} + g_{ds2} + g_{m2}) + C_2(g_{ds2} + G_L) + C_1(g_{ds2} + G_L)} \approx \frac{-G_L}{C_3}$

The nondominant root p_2 is given as $p_2 = \frac{-[C_3(g_{ds1} + g_{ds2} + g_{m2}) + C_2(g_{ds2} + G_L) + C_1(g_{ds2} + G_L)]}{C_3(C_1 + C_2)} \approx \frac{-g_{m2}}{C_1 + C_2}$

Assuming that C_1 , C_2 , and C_3 are the same order of magnitude, and that g_{m2} is greater than G_L , then $|p_1|$ is smaller than $|p_2|$ (closer to the origin). Therefore, the approximation of $|p_2| \gg |p_1|$ is valid.

Notice that there is a right-half plane zero at $z_1 = g_{m1}/C_1$.



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Example: Cascode Frequency Response

Calculate the -3 dB frequency of the cascode voltage amplifier, which has the following parameters:

$$R_I = 10 \text{ k}\Omega$$

$$R_L = 10 \text{ k}\Omega$$

$$C_{gs1} = 5 \text{ pF}$$

$$C_{gs2} = 10 \text{ pF}$$

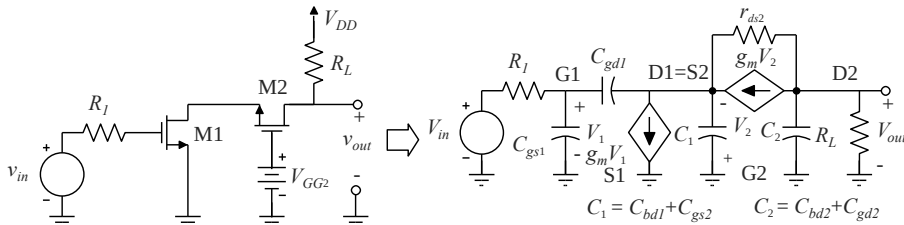
$$C_{gd1} = C_{gd2} = 1 \text{ pF}$$

$$C_{bd1} = C_{bd2} = 2 \text{ pF}$$

$$g_{m1} = 3 \text{ mA/V}$$

$$g_{m2} = 6 \text{ mA/V}$$

$$r_{ds2} = 50 \text{ k}\Omega$$



Example schematic and equivalent model

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Example: Cascode Frequency Response (continued)

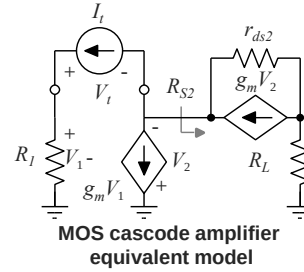
Find the open-circuit time constants:

$$C_{gs1}: R_{gs01} = R_I = 10k\Omega$$

$$C_{gd1}: R_{gd01} = ?$$

$$V_t = I_t R_I + (I_t + g_{m1} V_1) R_{S2} = I_t R_I + (I_t + g_{m1} I_t R_{I1}) R_{S2}$$

$$\therefore R_{gd01} = \frac{V_t}{I_t} = R_I + R_{S2} + g_{m1} R_I R_{S2} = R_I + R_{S2} (1 + g_{m1} R_I)$$



But what is R_{S2} ? Using the model shown, you can see that

$$I_t = g_{m2} V_t \left(\frac{r_{ds2}}{r_{ds2} + R_{L2}} \right) \Rightarrow R_{S2} = \frac{V_t}{I_t} = \frac{r_{ds2} + R_{L2}}{g_{m2} r_{ds2}}$$

Notice that unless $r_{ds2} > R_{L2}$, then R_{S2} is greater than $1/g_{m2}$.

$$\therefore R_{gd01} = R_I + (1 + g_{m1} R_I) \left(\frac{r_{ds2} + R_{L2}}{g_{m2} r_{ds2}} \right) = 10k\Omega + 31 \cdot 0.2k\Omega = 16.2k\Omega$$

$$C_1: R_{01} = R_{S2} = \frac{r_{ds2} + R_{L2}}{g_{m2} r_{ds2}} = 0.2k\Omega \quad C_2: R_{01} = R_{L2} = 10k\Omega$$

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Example: Cascode Frequency Response (continued)

$$\begin{aligned}
 \omega_{-3dB} &\approx \frac{1}{\sum T_0} = \frac{1}{R_{gs01}C_{gs1} + R_{gd01}C_{gd1} + R_{01}C_1 + R_{02}C_2} \\
 &= \frac{1}{R_{gs01}C_{gs1} + R_{gd01}C_{gd1} + R_{01}(C_{bd1} + C_{gs2}) + R_{02}(C_{bd2} + C_{gd2})} \\
 &= \frac{10^9}{10 \cdot 5 + 16.2 \cdot 1 + 0.2 \cdot 12 + 10 \cdot 3} = \frac{10^9}{64.8} = 15.43 \times 10^6 \text{ rads/s} \rightarrow f_{-3dB} = 2.46 \text{ MHz}
 \end{aligned}$$

Note: The Miller effect, R_{gd01} , is less in the cascode amplifier (16.2 kΩ, compared with 320 kΩ).



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Dominant Pole Approximation

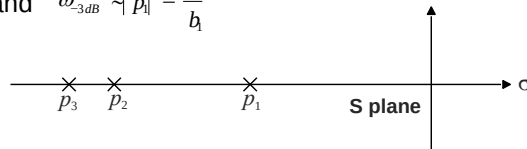
If one of the poles is significantly closer to the origin of the complex frequency plane, its magnitude is a good approximation to the -3 dB frequency. Consider the following general transfer function:

$$A(s) = \frac{N(s)}{D(s)} = \frac{a_0 + a_1s + a_2s^2 + \dots + a_ms^m}{b_0 + b_1s + b_2s^2 + \dots + b_ns^n} = \frac{K}{\left(1 - \frac{s}{p_1}\right) \left(1 - \frac{s}{p_2}\right) \dots \left(1 - \frac{s}{p_n}\right)}$$

Equating denominator terms gives

$$b_1 = -\frac{1}{p_1} - \frac{1}{p_2} - \dots - \frac{1}{p_n} = \sum_{i=1}^n \left(-\frac{1}{p_i} \right)$$

$$\text{If } |p_1| \ll |p_2|, |p_3|, \dots \text{ then } b_1 \approx -\frac{1}{p_1} \text{ and } \omega_{-3dB} \approx |p_1| = \frac{1}{b_1}$$



Plot of poles on the complex plane



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Open-Circuit Time-Constant Analysis

- The open-circuit time-constant analysis is a method suitable for finding the dominant pole of a circuit with multiple capacitors.
- Consider the following n-port network. You can express the nodal equations as:

$$I_1 = (g_{11} + sC_1)V_1 + g_{12}V_2 + g_{13}V_3 + \dots$$

$$I_2 = g_{21}V_1 + (g_{22} + sC_2)V_2 + g_{23}V_3 + \dots$$

$$I_3 = g_{31}V_1 + g_{32}V_2 + (g_{33} + sC_3)V_3 + \dots$$

- The determinant of the previous can be expressed as:

$$\Delta(s) = K_0 + K_1s + K_2s^2 + K_3s^3 = K_0(1 + b_1s + b_2s^2 + b_3s^3)$$

Where:

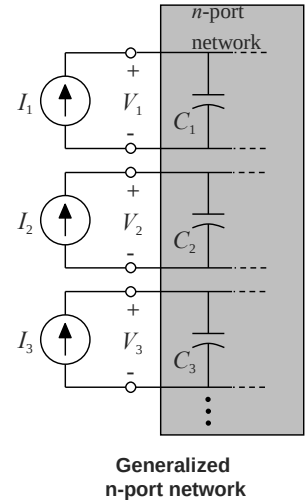
$$K_0 = \Delta(C_i = 0) \text{ for all } i \neq 0$$

- Consider now the K_1 which involves a single capacitor, and is given as:

$$K_1 = h_1sC_1 + h_2sC_2 + h_3sC_3$$

- The h_i terms can be evaluated by expanding the determinant about each row:

$$\text{First row: } \Delta(s) = (g_{11} + sC_1)\Delta_{11} + g_{12}\Delta_{12} + g_{13}\Delta_{13} \Rightarrow h_1 = \Delta_{11}(C_i = 0, i \neq 1)$$



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Open-Circuit Time-Constant Analysis (continued)

Second row: $\Delta(s) = g_{21}\Delta_{21} + (g_{22} + sC_2)\Delta_{22} + g_{23}\Delta_{23} \Rightarrow h_2 = \Delta_{22}(C_i = 0, i \neq 2)$

Third row: $\Delta(s) = g_{31}\Delta_{31} + g_{32}\Delta_{32} + (g_{33} + sC_3)\Delta_{33} \Rightarrow h_3 = \Delta_{33}(C_i = 0, i \neq 3)$

$$\therefore K_1 = C_1\Delta_{11}(C_i = 0, i \neq 1) + C_2\Delta_{22}(C_i = 0, i \neq 1) + C_3\Delta_{33}(C_i = 0, i \neq 3)$$

Finally:

$$b_1 = \frac{K_1}{K_0} = \frac{\Delta_{11}(C_i = 0, i \neq 1)}{\Delta_0} C_1 + \frac{\Delta_{22}(C_i = 0, i \neq 2)}{\Delta_0} C_2 + \frac{\Delta_{33}(C_i = 0, i \neq 3)}{\Delta_0} C_3$$

If the driving-point impedance of the i -th port is expressed as $\frac{V_i}{I_i} = \frac{\Delta_{11}}{\Delta(s)}$,
then $R_{1o} = \frac{\Delta_{11}(C_i = 0, i \neq 1)}{\Delta_0}$, $R_{2o} = \frac{\Delta_{22}(C_i = 0, i \neq 2)}{\Delta_0}$, $R_{3o} = \frac{\Delta_{33}(C_i = 0, i \neq 3)}{\Delta_0}, \dots$

are the driving-point impedances at ports 1, 2, 3, ... with all capacitors set equal to zero.

$$\therefore b_1 = R_{1o}C_1 + R_{2o}C_2 + R_{3o}C_3 + \dots$$

where $R_{io}C_i$ are called the *open-circuit time constants*.

If there are no dominant zeros, then the dominant pole, p_1 , is given as:

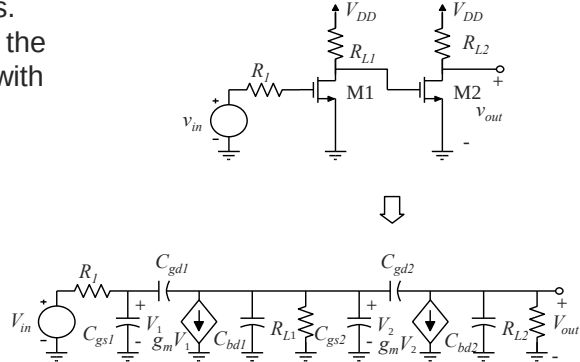
$$\omega_{-3dB} \approx |p_1| = \frac{1}{b_1} = \frac{1}{R_{1o}C_1 + R_{2o}C_2 + R_{3o}C_3 + \dots} = \frac{1}{\sum_{i=1}^n R_{io}C_i}$$



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MOS Cascade Amplifier

The MOS cascade amplifier consists of two common-source stages connected in series. The drain of the first FET drives the gate of the second FET. An example is shown, along with its equivalent model.



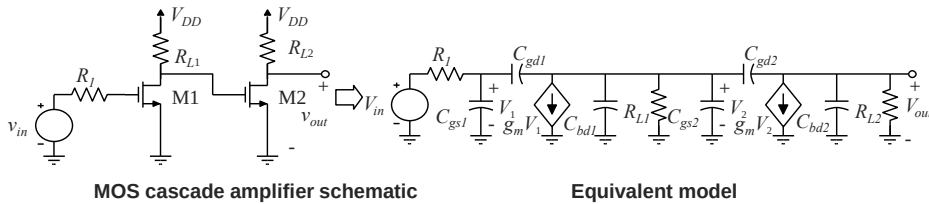
MOS cascade amplifier schematic and equivalent model

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Example: Cascade Frequency Response

Calculate the -3 dB frequency of this cascade voltage amplifier, which has the following parameters:

$$\begin{array}{lll}
 R_I = 10 \text{ k}\Omega & R_{L1} = 10 \text{ k}\Omega & R_{L2} = 5 \text{ k}\Omega \\
 C_{gs1} = 5 \text{ pF} & C_{gs2} = 10 \text{ pF} & C_{gd1} = C_{gd2} = 1 \text{ pF} \\
 C_{bd1} = C_{bd2} = 2 \text{ pF} & g_{m1} = 3 \text{ mA/V} & g_{m2} = 6 \text{ mA/V}
 \end{array}$$



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Example: Cascade Frequency Response (continued)

Find the open-circuit time constants:

$$C_{gs1} \cdot R_{gs01} = R_I = 10k\Omega \quad C_{gs2} \cdot R_{gs02} = R_{L1} = 10k\Omega$$

C_{gd1} : (Use the model to the right)

$$V_t = I_t R_I + (I_t + g_{m1} V_1) R_{L1} = I_t R + (I_t + g_{m1} I_t R_I) R_{L1}$$

$$\therefore R_{gd01} = \frac{V_t}{I_t} = R_I + R_{L1} + g_{m1} R_I R_{L1} = 20k\Omega + 300k\Omega = 320k\Omega$$

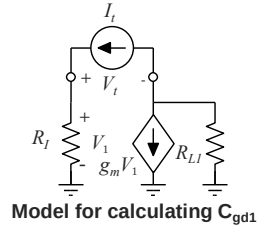
C_{gd2} : (You can use the results of C_{gd1})

$$\therefore R_{gd02} = R_{L1} + R_{L2} + g_{m2} R_{L1} R_{L2} = 10k\Omega + 5k\Omega + 300k\Omega = 315k\Omega$$

$$C_{bd1}: R_{bd01} = R_{L1} = 10k\Omega \quad C_{bd2}: R_{bd02} = R_{L2} = 5k\Omega$$

$$\omega_{-3dB} \approx \frac{1}{\sum T_0} = \frac{1}{R_{gs01} C_{gs1} + R_{gs02} C_{gs2} + R_{gd01} C_{gs1} + R_{gs02} C_{gs2} + R_{bd01} C_{bd1} + R_{bd02} C_{bd2}}$$

$$= \frac{10^9}{10 \cdot 5 + 10 \cdot 10 + 320 \cdot 1 + 315 \cdot 1 + 10 \cdot 2 + 5 \cdot 2} = \frac{10^9}{815} = 1.227 \times 10^6 \text{ rads/s} \rightarrow f_{-3dB} = 195 \text{ kHz}$$



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Example: Cascade Frequency Response (continued)

- Computer simulation gives poles at -205 kHz, -4.02 MHz, and -39.98 MHz and two zeros at $+477$ MHz and $+955$ MHz.
- How important is it for the circuit to have a dominant pole for the open-circuit time-constant approach?
- For a circuit with two identical poles, the -3 dB frequency is:

$$\omega_{-3dB} = \omega_x \sqrt{\sqrt{2} - 1} = 0.64\omega_x$$

- The open-circuit time-constant approach gives

$$\omega_{-3dB} = \omega_x / 2 = 0.5\omega_x \rightarrow 22\% \text{ error and is pessimistic.}$$



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Simple Two-Stage Amplifier

DC Conditions

$$I_5 = I_{bias}, I_1 = I_2 = 0.5I_5 = 0.5I_{bias}, I_7 = I_6 = nI_{bias}$$

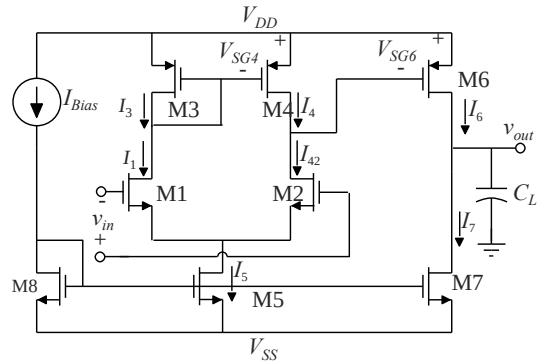
$$V_{icm}(\max) = V_{DD} - V_{SG3} + V_{T1}$$

$$V_{icm}(\min) = V_{SS} + V_{DS5(sat)} + V_{GS1}$$

$$V_{out}(\max) = V_{DD} - V_{SD6(sat)}$$

$$V_{out}(\min) = V_{SS} + V_{DS7(sat)}$$

Determined by triode crash



Notice that the output stage is class A:

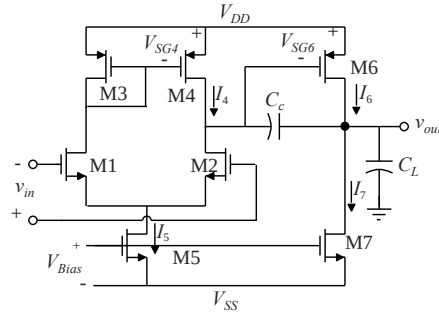
$$\therefore I_{\text{sink}} = I_7$$

$$I_{\text{source}} = \frac{K'_N W_6}{2L_6} (V_{DD} - V_{SS} - V_T)^2 - I_7$$

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Two-Stage Amplifier: DC Balance Conditions

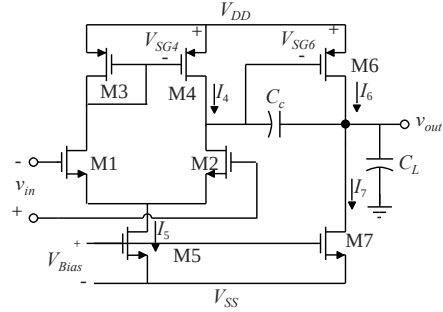
- For best performance, keep all transistors in saturation.
- M4 is the only transistor that cannot be forced into saturation by internal connections or external voltages.
- Therefore, develop conditions to force M4 to be in saturation.



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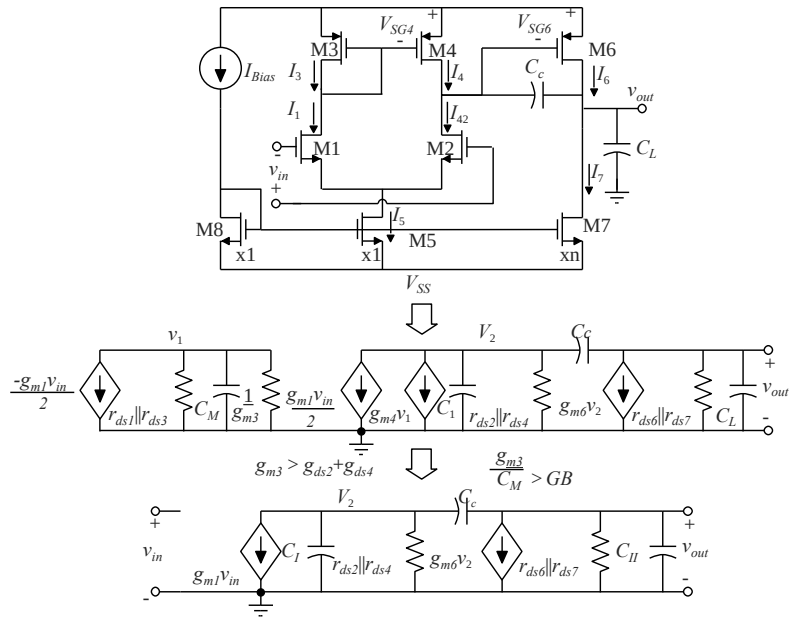
DC Balance Conditions (continued)

1. Assume that $V_{SG4} = V_{SG6}$, which causes proper mirroring in the M3-M4 mirror. Also, the gate and drain of M4 are at the same potential so that M4 is guaranteed to be in saturation.
2. Let $S_i = \frac{W_i}{L_i}$, if $V_{SG4} = V_{SG6}$ then $I_6 = \left(\frac{S_6}{S_4}\right)I_4$
3. However, $I_7 = \left(\frac{S_7}{S_5}\right)I_5 = \left(\frac{S_7}{S_5}\right)(2I_4)$
4. For balance, I_6 must equal $I_7 \Rightarrow \frac{S_6}{S_4} = \frac{2S_7}{S_5}$
These are called *balance conditions*.
5. If the balance conditions are satisfied, then $V_{DG4} = 0$ and M4 is saturated.



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Two-Stage Amplifier: Equivalent Model



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Two-Stage Amplifier

Midband performance

$$A_0 = g_{m1} g_{mII} R_I R_{II} \approx g_{m1} g_{m6} (r_{ds2} \parallel r_{ds4}) (r_{ds6} \parallel r_{ds7}), \quad R_{out} = r_{ds6} \parallel r_{ds7}, \quad R_{in} = \infty$$

Zero at $z_1 = \frac{g_{mII}}{C_c} = \frac{g_{m6}}{C_c}$

Poles at $p_1 \approx \frac{-1}{g_{mII} R_I R_{II} C_c} = \frac{(g_{ds2} + g_{ds4})(g_{ds6} + g_{ds7})}{g_{m6} C_c}$ and $p_2 \approx \frac{-g_{mII}}{C_{II}} \approx \frac{-g_{m6}}{C_L}$

Assume that $g_{m1} = 100 \mu S$, $g_{m6} = 1 mS$, $r_{ds2} = r_{ds4} = 2 M\Omega$, $r_{ds6} = r_{ds7} = 0.5 M\Omega$, $C_c = 5 pF$, and $C_L = 10 pF$:

$$A_0 = (100 \mu S)(1 M\Omega)(1000 \mu S)(0.25 M\Omega) = 25000, \quad R_{in} = \infty, \quad R_{out} = 150 k\Omega$$

$$z_1 = \frac{1000 \mu S}{5 pF} = 2 \times 10^8 \text{ rads/sec or } 31.83 \text{ MHz}$$

$$p_1 = \frac{-1}{(1 mS)(1 M\Omega)(0.25 M\Omega)(5 pF)} = -800 \text{ rads/sec or } 127.3 \text{ Hz}$$

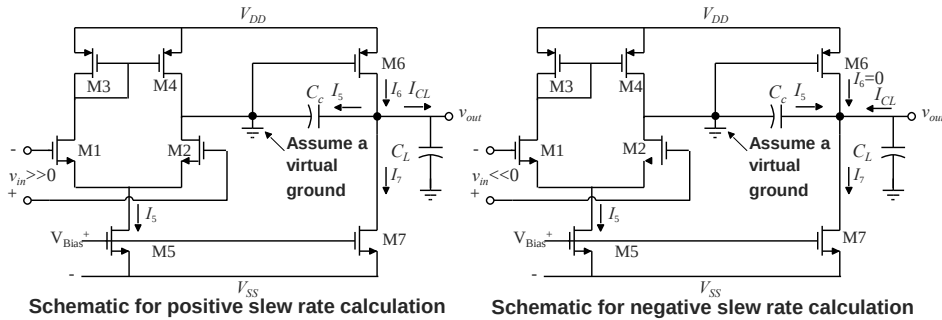
$$GB = 3.178 \text{ MHz}$$

$$p_2 = \frac{-1000 \mu S}{10 pF} = 10^8 \text{ rads/sec} = 15.92 \text{ MHz}$$



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Two-Stage Amplifier: Slew Rate



Remember that the slew rate occurs when currents flowing in a capacitor become limited and is given as $I_{\text{lim}} = C \frac{dv_C}{dt}$

where v_C is the voltage across the capacitor C .

$$SR^+ = \min \left[\frac{I_5}{C_C}, \frac{I_6 - I_5 - I_7}{C_L} \right] = \frac{I_5}{C_C} \text{ because } I_6 \gg I_5$$

$$SR^- = \min \left[\frac{I_5}{C_C}, \frac{I_7 - I_5}{C_L} \right] = \frac{I_5}{C_C} \text{ if } I_7 \gg I_5$$

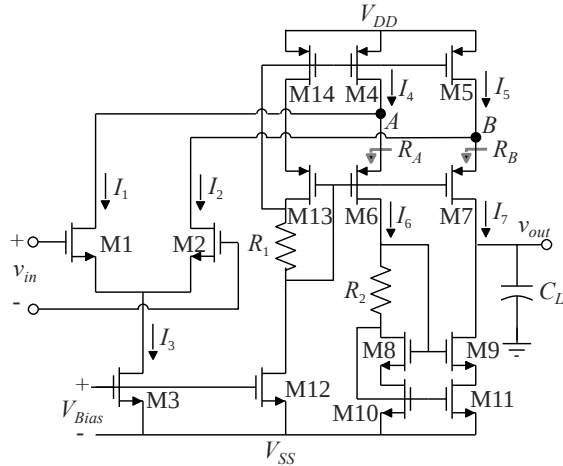
Therefore, if C_L is not too large and if I_7 is significantly greater than I_5 , then the slew rate of the two-stage opamp should be $SR = \frac{I_5}{C_C}$.

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Folded-Cascode Multistage Amplifier

The folded-cascode CMOS amplifier

- Has bias currents, I_4 and I_5 , that should be designed so that I_6 and I_7 never become zero (that is, $I_4 = I_5 = 1.5I_3$).
- Is nearly balanced (would be exactly if R_A was equal to R_B).
- Is self-compensating.
- Has poor noise performance. The gain occurs at the output, so all intermediate transistors contribute to the noise along with the input transistors. (Some first-stage gain can be achieved if R_A and R_B are greater than g_{m1} or g_{m2}).

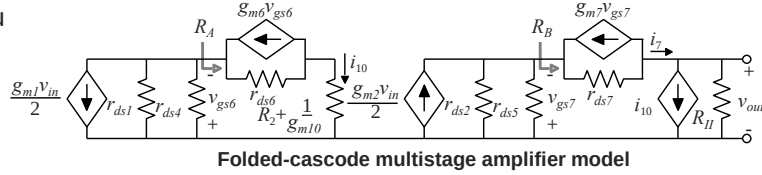


Folded-cascode multistage amplifier schematic

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Small-Signal Analysis of the Folded Cascode Amplifier

Recalling what you learned about the resistance looking into the source of the cascode transistor gives:



$$R_A = \frac{r_{ds6} + R_2 + (1/g_{m10})}{1 + g_{m6}r_{ds6}} \approx \frac{1}{g_{m6}} \text{ and } R_B = \frac{r_{ds7} + R_{II}}{1 + g_{m7}r_{ds7}} \approx \frac{R_{II}}{g_{m7}r_{ds7}} \text{ where } R_{II} \approx g_{m9}r_{ds9}r_{ds11}$$

The small-signal voltage transfer function can be found as follows. The current i_{10} is

$$\text{written as } i_{10} = \frac{-g_{m1}(r_{ds1} \parallel r_{ds4})v_{in}}{2[R_A + (r_{ds1} \parallel r_{ds4})]} \approx \frac{-g_{m1}v_{in}}{2}$$

and the current i_7 can be expressed as:

$$i_7 = \frac{\frac{g_{m2}(r_{ds2} \parallel r_{ds5})v_{in}}{2 \left[\frac{R_{II}}{g_{m7}r_{ds7}} + (r_{ds2} \parallel r_{ds5}) \right]}}{2 \left(1 + \frac{R_{II}(g_{ds2} + g_{ds5})}{g_{m7}r_{ds7}} \right)} = \frac{g_{m2}v_{in}}{2(1+k)} \text{ where } k = \frac{R_{II}(g_{ds2} + g_{ds5})}{g_{m7}r_{ds7}}$$

The output voltage, v_{out} , is equal to the sum of i_7 and i_{10} flowing through R_{out} . Thus:

$$\frac{v_{out}}{v_{in}} = \left(\frac{g_{m1}}{2} + \frac{g_{m2}}{2(1+k)} \right) R_{out} = \left(\frac{2+k}{2+2k} \right) g_{m1} R_{out}$$

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Frequency Response: Folded Cascode

The frequency response of the folded cascode opamp is determined primarily by the output pole which is given as

$$p_{out} = \frac{-1}{R_{out}C_{out}}$$

where C_{out} is all the capacitance connected from the output of the opamp to ground. All other poles must be greater than $GB = g_{m1}/C_{out}$. The approximate expressions for each pole is described in the following:

Pole at node A:	$p_A \approx -1/R_A C_A$	Pole at source of M8:	$p_8 \approx -g_{m8}/C_8$
Pole at node B:	$p_B \approx -1/R_B C_B$	Pole at source of M9:	$p_9 \approx -g_{m9}/C_9$
Pole at drain of M6:	$p_6 \approx \frac{-1}{(R_2 + 1/g_{m10})C_6}$	Pole at gate of M10:	$p_{10} \approx -g_{m10}/C_{10}$

where the approximate expressions are found by the reciprocal product of the resistance and parasitic capacitance seen to ground from a given node. One might feel that because R_B is approximately r_{ds} that this pole might be too small. However, at frequencies where this pole has influence, C_{out} causes R_{out} to be much smaller making p_B also nondominant.



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Example: Folded-Cascode Multistage Amplifier

Assume that all $g_{mN} = g_{mP} = 100 \mu\text{S}$, $r_{dsN} = 2 \text{ M}\Omega$, $r_{dsP} = 1 \text{ M}\Omega$, and $C_L = 10 \text{ pF}$. Find all of the small-signal performance values for the folded-cascode opamp:

$$R_{II} = 0.4 \text{ G}\Omega, R_A = 10 \text{ k}\Omega, \text{ and } R_B = 4 \text{ M}\Omega \quad \therefore k = \frac{0.4 \times 10^9 (0.3 \times 10^{-6})}{100} = 1.2$$

$$\frac{v_{out}}{v_{in}} = \left(\frac{2+1.2}{2+2.4} \right) (100)(57.143) = (0.729)(5714.3) = 4,156$$

$$R_{out} = R_{II} \parallel [g_{m7} r_{ds7} (r_{ds5} \parallel r_{ds2})] = 400 \text{ M}\Omega \parallel [(100)(0.667 \text{ M}\Omega)] = 57.143 \text{ M}\Omega$$

$$|p_{out}| = \frac{1}{R_{out} C_{out}} = \frac{1}{(57.143 \text{ M}\Omega)(10 \text{ pF})} = 1750 \text{ rads/sec} \Rightarrow 278 \text{ Hz} \Rightarrow GB = 1.21 \text{ MHz}$$

- Folded cascodes provide good PSRR and ICMR and are self-compensated.
- You can cascade an output stage to get extremely high gain with lower output resistance (use Miller compensation in this case).
- You need first-stage gain for good noise performance.
- Folded cascodes are widely used in telecommunication circuits where large dynamic range is required.

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Output Stages

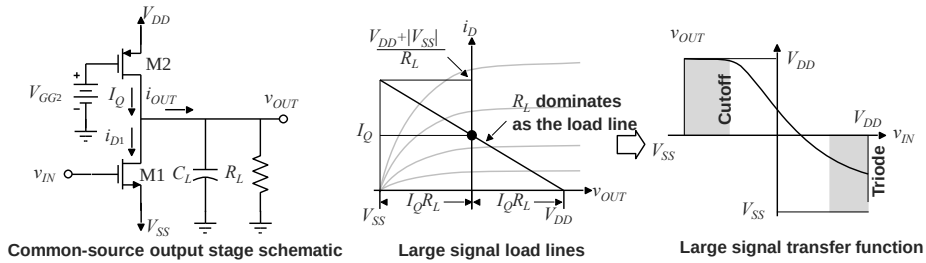
Why are output stages needed?

- **Load driving:** Some loads are capacitive and require more current from the driving stage.
- **Power efficiency:** In some cases, more efficient classes can be designed so that power efficiency is improved.
- **Low distortion:** Some applications require large-signal swings or minimal distortion.



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Common-Source Output Stage



A class A circuit has current flow in the MOSFETs during the entire period of a sinusoidal signal.

Characteristics of class A amplifiers

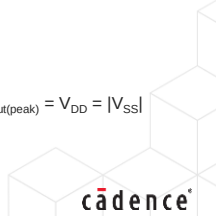
- Unsymmetrical sinking and sourcing of current
- Linear
- Poor efficiency

$$\frac{P_{RL}}{P_{vdd}} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD} - V_{SS})I_Q} = \frac{\frac{v_{out(peak)}^2}{2R_L}}{(V_{DD} - V_{SS}) \left(\frac{(V_{DD} - V_{SS})}{2R_L} \right)} = \left(\frac{v_{out(peak)}}{V_{DD} - V_{SS}} \right)^2$$

Efficiency = Maximum efficiency occurs when $v_{out(peak)} = V_{DD} - |V_{SS}|$ which gives 25%

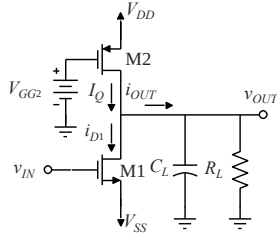
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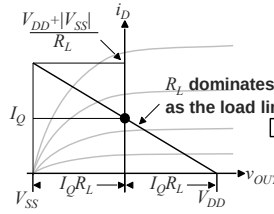


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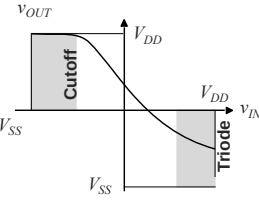
Common-Source Output Stage Performance



Common-source output stage schematic



Large-signal load lines



Large-signal transfer function

Output resistance:
$$r_{out} = \frac{1}{g_{ds1} + g_{ds2}} = \frac{1}{(\lambda_1 + \lambda_2)I_D}$$

Maximum sinking current:
$$I_{OUT}^- = \frac{K_1' W_1}{2L_1} (V_{DD} - V_{SS} - V_{T1})^2 - I_Q$$

Maximum sourcing current:
$$I_{OUT}^+ = \frac{K_2' W_2}{2L_2} (V_{DD} - V_{GG2} - |V_{T2}|)^2 \leq I_Q$$

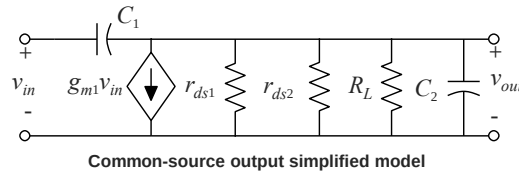
Requirements:
$$r_{out} \ll R_L \quad |I_{OUT}| > C_L \cdot SR \quad |I_{OUT}| > \frac{v_{OUT}(peak)}{R_L}$$

The maximum current is determined by *both* the current that is required to provide the necessary slew rate (C_L) and the current that is required to provide a voltage across the load resistor (R_L).

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Small-Signal Performance: Common-Source Output Stage

- Although you have considered the small-signal performance of the current source load inverter, you can include the influence of the load:



- The small-signal voltage gain is:
$$\frac{v_{out}}{v_{in}} = \frac{-g_{m1}}{g_{ds1} + g_{ds2} + G_L}$$
- The small-signal frequency response includes a:
 - Zero at $z = \frac{g_{m1}}{C_{gd1}}$
 - Pole at $p = \frac{-(g_{ds1} + g_{ds2} + G_L)}{C_{gd1} + C_{gd2} + C_{bd1} + C_{bd2} + C_L}$

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Example: Design of Class A Output Stage

Use the values of $K_N' = 110 \mu\text{A/V}^2$, $K_P' = 50 \mu\text{A/V}^2$, $V_{TN} = 0.7\text{V}$ and $V_{TP} = -0.7\text{V}$ and design the W/L ratios of M1 and M2 so that a voltage swing of ± 2 volts and a slew rate of $1\text{V}/\mu\text{s}$ is achieved if $R_L = 20 \text{ k}\Omega$ and $C_L = 1000 \text{ pF}$. Assume that $V_{DD} = |V_{SS}| = 3 \text{ volts}$ and $V_{GG2} = 0 \text{ volts}$. Let the channel lengths be $2 \mu\text{m}$ and assume that $C_{gd1} = 100 \text{ fF}$.

Solution

- Consider the effects of R_L and C_L :

$$i_{OUT}(\text{peak}) = \frac{\pm 2\text{V}}{20\text{k}\Omega} = \pm 100 \mu\text{A} \text{ and } C_L \cdot SR = 10^{-9} \cdot 10^6 = 1000 \mu\text{A}$$

- Because the slew rate current $\gg$ the current for R_L , you can safely assume that all of the current supplied by the inverter is available to charge C_L . Using a value of $\pm 1 \text{ mA}$:

$$\frac{W_1}{L_1} = \frac{2(I_{OUT}^- + I_Q)}{K_N'(V_{DD} + |V_{SS}| - V_{TN})^2} = \frac{4000}{110 \cdot (5.3)^2} = \frac{3 \mu\text{m}}{2 \mu\text{m}} \text{ and } \frac{W_2}{L_2} = \frac{2I_{OUT}^+}{K_P'(V_{DD} - V_{GG2} - |V_{TP}|)^2} = \frac{2000}{50 \cdot (2.3)^2} \approx \frac{15 \mu\text{m}}{2 \mu\text{m}}$$

- The small-signal performance of this amplifier is $A_v = -8.21 \text{ V/V}$ (includes $R_L = 20 \text{ k}\Omega$).



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Broadband Harmonic Distortion

- The linearity of an amplifier can be characterized by its influence on a pure sinusoidal input signal:
- Assume the input is $V_{in}(\omega) = V_p \sin(\omega t)$
- The output of an amplifier with distortion is $V_{out}(\omega) = a_1 V_p \sin(\omega t) + a_2 V_p \sin(2\omega t) + \dots + a_n V_p \sin(n\omega t)$
- *Harmonic distortion* (HD) for the i-th harmonic can be defined as the ratio of the magnitude of the i-th harmonic to the magnitude of the fundamental.
- For example, second-harmonic distortion is given as $HD_2 = \frac{a_2}{a_1}$



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Broadband Harmonic Distortion (continued)

- *Total harmonic distortion* (THD) is defined as the square root of the ratio of the sum of all of the second and higher harmonics to the magnitude of the first or fundamental harmonic. Thus, THD can be expressed as:

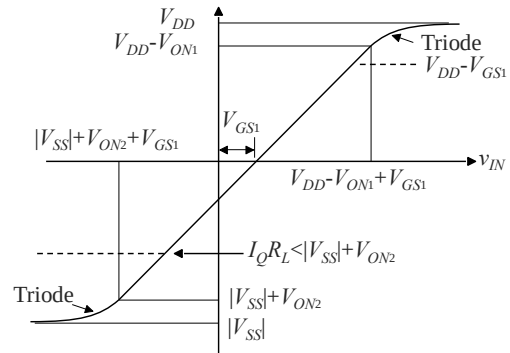
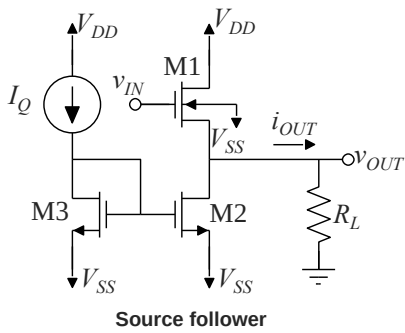
$$THD = \frac{[a_2^2 + a_3^2 + \dots + a_n^2]^{1/2}}{a_1}$$

- The distortion of the class A amplifier is good for small signals and becomes poor at maximum-output swings because of the nonlinearity of the voltage-transfer curve for large-signal swing.



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Source Follower



Voltage-transfer curve with current sink bias

Maximum-output voltage swings:

$$v_{OUT}(\min) \approx V_{SS} - V_{ON2} \text{ (if } R_L \text{ is large) or } v_{OUT}(\min) \approx -I_Q R_L \text{ (if } R_L \text{ is small)}$$

$$v_{OUT}(\max) = V_{DD} - V_{ON1} \text{ (if } v_{IN} \gg V_{DD}) \text{ or } v_{OUT}(\max) \approx V_{DD} - V_{GS1}$$

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Source Follower: Output-Voltage Swing

- The previous results do not include the bulk effect on V_{T1} of V_{GS1} . Therefore:

$$V_{T1} = V_{T01} + \gamma \left[\sqrt{2|\phi_F| - v_{BS}} - \sqrt{2|\phi_F|} \right] \approx V_{T01} + \gamma \sqrt{v_{SB}} = V_{T01} + \gamma_1 \sqrt{v_{OUT}(\max) - V_{SS}}$$

$$\therefore v_{OUT}(\max) - V_{SS} \approx V_{DD} - V_{SS} - V_{ON1} - V_{T1} = V_{DD} - V_{SS} - V_{ON1} - V_{T01} - \gamma_1 \sqrt{v_{OUT}(\max) - V_{SS}}$$

- Define $v_{OUT}(\max) - V_{SS} = v_{OUT}(\max)$, which gives the quadratic:

$$v_{OUT}(\max) + \gamma_1 \sqrt{v_{OUT}(\max)} - (V_{DD} - V_{SS} - V_{ON1} - V_{T01}) = 0$$

- Solving the quadratic gives:

$$v_{OUT}(\max) \approx \frac{\gamma_1^2}{4} - \frac{\gamma_1}{2} \sqrt{\gamma_1^2 + 4(V_{DD} - V_{SS} - V_{ON1} - V_{T01})} + \frac{\gamma_1^2 4(V_{DD} - V_{SS} - V_{ON1} - V_{T01})}{4}$$

If $V_{DD} = 2.5V$, $\gamma_N = 0.4V^{1/2}$, $V_{TN1} = 0.7V$, and $V_{ON1} = 0.2V$, then $v_{OUT}(\max) = 3.661V$ and $v_{OUT}(\max) = 3.661 - 2.5 = 0.861V$.



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Maximum Sourcing and Sinking Currents

Maximum sourcing current (in a short circuit):

Assume that the transistors are in saturation and $V_{DD} = -V_{SS} = 2.5\text{V}$ and thus

$$I_{OUT}(\text{sourcing}) = \frac{K_1' W_1}{2L_1} [V_{DD} - v_{OUT} - V_{T1}]^2 - I_Q$$

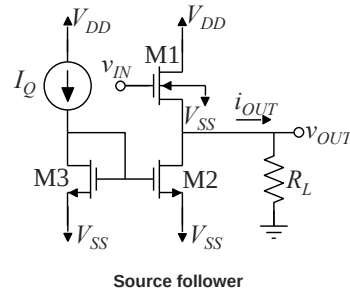
where v_{IN} is assumed to be equal to V_{DD} .

If $W_1/L_1 = 10$ and if $v_{OUT} = 0\text{V}$, then $V_{T1} = 1.08\text{V} \Rightarrow I_{OUT}$ equal to 1.11 mA.

However, as v_{OUT} increases above 0V, the current rapidly decreases.

Maximum sinking current: For the current sink load, the sinking current is whatever the sink is biased to provide

$$I_{OUT}(\text{sinking}) = I_Q$$

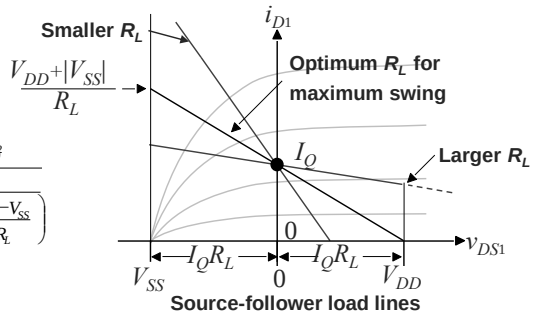


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Source-Follower Efficiency

Assume that the source follower can swing to power supply.

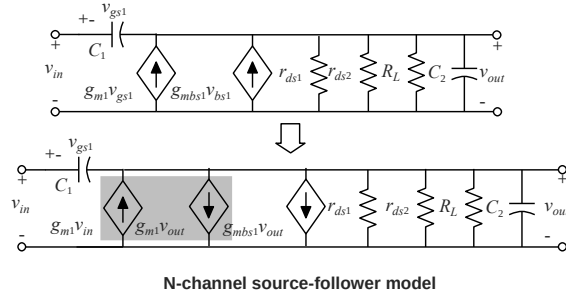
$$\begin{aligned}\text{Efficiency} &= \frac{P_{RL}}{P_{\text{Supply}}} = \frac{v_{OUT}(\text{peak})}{\frac{2R_L}{(V_{DD} - V_{SS})I_Q}} = \frac{v_{OUT}(\text{peak})}{(V_{DD} - V_{SS}) \left(\frac{V_{DD} - V_{SS}}{2R_L} \right)} \\ &= \left(\frac{v_{OUT}(\text{peak})}{V_{DD} - V_{SS}} \right)^2\end{aligned}$$



- Maximum efficiency occurs when $v_{OUT}(\text{peak}) = V_{DD} - |V_{SS}|$ which gives 25%.
- Maximum efficiency occurs for the optimum value of R_L , which gives maximum swing.
- Other values of R_L result in less efficiency (and smaller signal swings before clipping).
- One fact has been ignored: The dynamic Q point *cannot* travel along the full length of the load line because of minimum and maximum voltage limits.

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Source-Follower Small-Signal Performance



$$\frac{V_{out}}{V_{in}} = \frac{g_{m1}}{g_{ds1} + g_{ds2} + g_{m1} + g_{mbs1} + G_L} \cong \frac{g_{m1}}{g_{m1} + g_{mbs1} + G_L} \cong \frac{g_{m1} R_L}{1 + g_{m1} R_L}$$

If $V_{DD} = -V_{SS} = 2.5V$, $V_{out} = 0V$, $W_1/L_1 = 10 \mu m/1 \mu m$, $W_2/L_2 = 1 \mu m/1 \mu m$, and $I_D = 500 \mu A$, then for the current sink load follower ($R_L = \infty$):

$$\frac{V_{out}}{V_{in}} = 0.869V/V \quad \text{If the bulk effect is ignored, then} \quad \frac{V_{out}}{V_{in}} = 0.963V/V$$

$$\text{For a finite load, } R_L = 1000\Omega: \frac{V_{out}}{V_{in}} = 0.512V/V$$

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Source-Follower Performance (continued)

The output resistance is $R_{out} = \frac{1}{g_{m1} + g_{mbs1} + g_{ds1} + g_{ds2} + G_L}$

For the current sink load follower: $R_{out} = 830\Omega$

The frequency response of the source follower:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{g_{m1} + sC_1}{g_{ds1} + g_{ds2} + g_{m1} + g_{mbs1} + G_L + s(C_1 + C_2)}$$

Where:

C_1 = capacitances connected between the input and output $\approx C_{GS1}$

$C_2 = C_{bs1} + C_{bd2} + C_{gd2}(\text{or } C_{gs2}) + C_L$

$$z = -\frac{g_{m1}}{C_1} \quad \text{Where: } p \approx -\frac{g_{m1} + C_L}{C_1 + C_2}$$

The presence of an LHP zero leads to the possibility that, in most cases, the pole and zero provide some degree of cancellation leading to a broadband response.

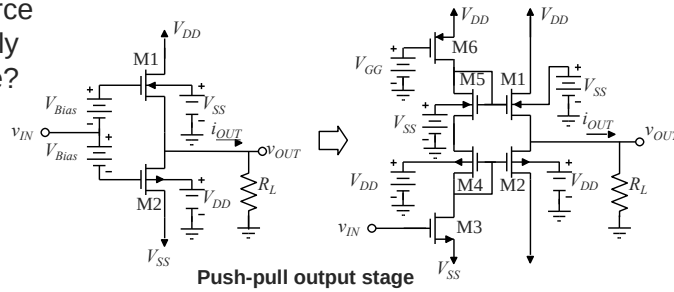


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Push-Pull MOS Output Stages: Classes AB and A

Can both sink and source current provide a slightly lower output resistance?

Efficiency depends on how the transistors are biased.



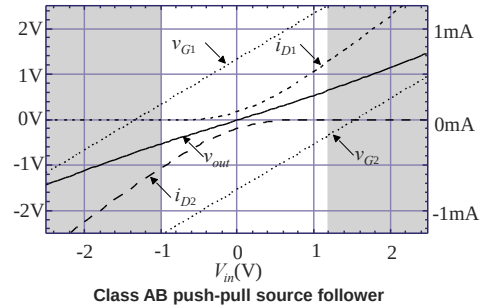
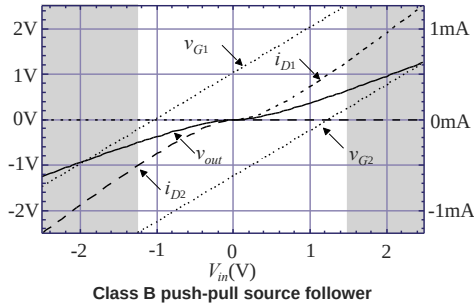
Class B: One transistor has current flow for only 180° of the sinusoid (half period).

$$\therefore \text{Efficiency} = \frac{P_{RL}}{P_{VDD}} = \frac{\frac{v_{OUT}(\text{peak})^2}{2R_L}}{(V_{DD} - V_{SS}) \left(\frac{1}{2} \right) \left(\frac{2v_{OUT}(\text{peak})}{\pi R_L} \right)} = \frac{\pi v_{OUT}(\text{peak})}{2 V_{DD} - V_{SS}}$$

Class AB: Each transistor has current flow for more than 180° of the sinusoid. Maximum efficiency is between 25% and 78.5%. Maximum efficiency occurs when $v_{OUT}(\text{peak}) = V_{DD}$ and is 78.5%.

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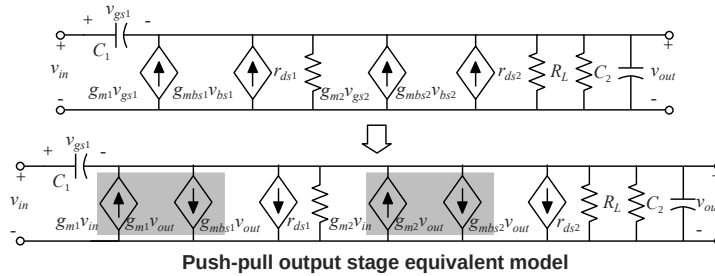
Class B and AB Push-Pull Source Followers



- Notice that v_{OUT} cannot reach the extreme values of V_{DD} and V_{SS} .
- $I_{OUT}^+(\text{max})$ and $I_{OUT}^-(\text{max})$ is always less than V_{DD}/R_L or V_{SS}/R_L .
- For $v_{OUT} = 0\text{V}$, quiescent current is flowing in M1 and M2 for class AB.
- Notice that there is significant distortion at $v_{IN} = 0\text{V}$ for the class B push-pull follower.

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Push-Pull Follower: Small-Signal Performance



$$\frac{v_{out}}{v_{in}} = \frac{g_{m1} + g_{m2}}{g_{ds1} + g_{ds2} + g_{m1} + g_{mbs1} + g_{m2} + g_{mbs2} + G_L} \quad R_{out} = \frac{1}{g_{ds1} + g_{ds2} + g_{m1} + g_{mbs1} + g_{m2} + g_{mbs2}} \text{ (does not include } R_L \text{)}$$

If $V_{DD} = -V_{SS} = 2.5V$, $V_{out} = 0V$, $I_{D1} = I_{D2} = 500 \mu A$, and $W/L = 20 \mu m/2 \mu m$, $A_v = 0.787$ ($R_L = \infty$) and $R_{out} = 448 \Omega$.

A zero and pole are located at:

$$z = \frac{-(g_{m1} + g_{m2})}{C_1} \quad p = \frac{-(g_{ds1} + g_{ds2} + g_{m1} + g_{mbs1} + g_{m2} + g_{mbs2} + G_L)}{C_1 + C_2}$$

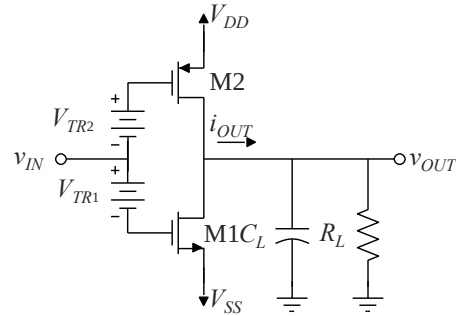
These roots are high frequency because the associated resistances are small.

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Push-Pull Common-Source Output Stage

Although similar to a class A amplifier, a push-pull common-source amplifier can operate as a class B amplifier to provide higher efficiency:

- The batteries V_{TR1} and V_{TR2} are necessary to control the bias current in M1 and M2.
- The efficiency is the same as the push-pull source follower.



Push-pull common-source output stage

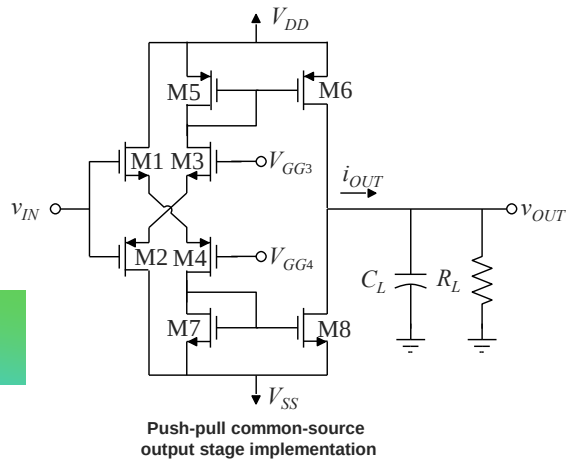
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Push-Pull Common-Source Output Stage: Method 1

One method to implement a push-pull common-source amplifier is as follows:

- V_{GG3} and V_{GG4} can be used to bias this amplifier in a class AB or class B operation.

Note: The bias current in M6 and M8 is *not* dependent upon V_{DD} or V_{SS} (assuming V_{GG3} and V_{GG4} are not dependent on V_{DD} and V_{SS}).

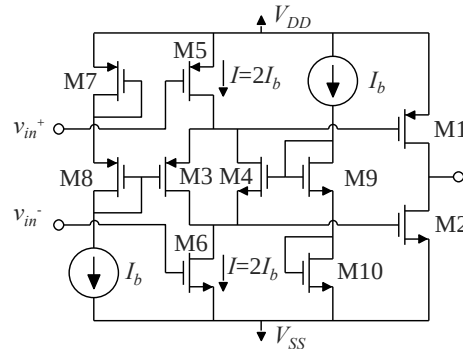


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Method 1 (continued)

In steady-state, the current through M5 and M6 is $2I_b$. If $W_4/L_4 = W_9/L_9$ and $W_3/L_3 = W_8/L_8$ then the currents in M1 and M2 can be determined by the following relationship:

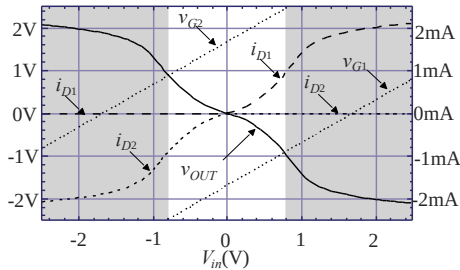
$$I_1 = I_2 = I_b \left(\frac{W_1/L_1}{W_7/L_7} \right) = I_b \left(\frac{W_2/L_2}{W_{10}/L_{10}} \right)$$



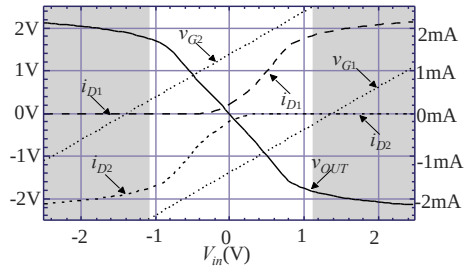
If v_{in+} goes low, M5 pulls the gates of M1 and M2 high. M4 shuts off, causing all of the current flowing through M5 ($2I_b$) to flow through M3, shutting off M1. The gate of M2 is high, allowing the buffer to strongly sink current. If v_{in-} goes high, M6 pulls the gates of M1 and M2 low. As before, this shuts off M2 and turns on M1, allowing strong sourcing.

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Classes B and AB Push-Pull Common-Source Amplifiers



Class B push-pull common-source amplifier



Class AB push-pull common-source amplifier

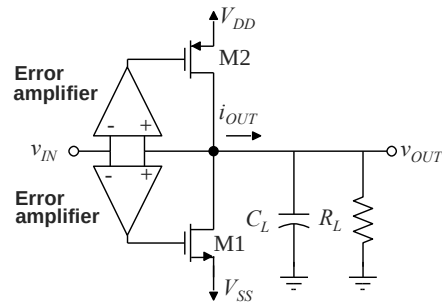
- There is significant distortion at $v_{IN} = 0V$ for the class B inverter.
- v_{OUT} cannot reach the extreme values of V_{DD} and V_{SS} .
- $I_{OUT}^+(\max)$ and $I_{OUT}^-(\max)$ is always less than V_{DD}/R_L or V_{SS}/R_L .
- For $v_{OUT} = 0V$, there is quiescent current flowing in M1 and M2 for class AB.

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Reducing Output Resistance: Negative Shunt Feedback

You can use negative shunt feedback to reduce output resistance, which:

- Can achieve output resistances as low as 10Ω .
- If the error amplifiers are not balanced, it is difficult to control the quiescent current in M1 and M2.
- Produces great linearity because of the strong feedback.
- Can be efficient if operated in class B or class AB.



$$R_{out} = \frac{r_{ds1} \parallel r_{ds2}}{1 + \text{loop gain}}$$

Negative shunt feedback example

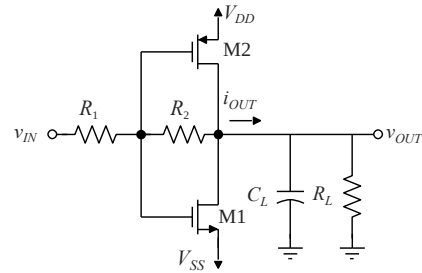


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Implementing Negative Shunt Feedback

$$\text{Loop gain} \approx \left(\frac{R_1}{R_1 + R_2} \right) \left(\frac{g_{m1} + g_{m2}}{g_{ds1} + g_{ds2} + G_L} \right)$$

$$\therefore R_{out} = \frac{r_{ds1} \| r_{ds2}}{1 + \left(\frac{R_1}{R_1 + R_2} \right) \left(\frac{g_{m1} + g_{m2}}{g_{ds1} + g_{ds2} + G_L} \right)}$$



Negative shunt feedback

Let $R_1 = R_2$, $R_L = \infty$, $I_{Bias} = 500 \mu A$, $W_1/L_1 = 100 \mu m/1 \mu m$ and $W_2/L_2 = 200 \mu m/1 \mu m$.

Thus, $g_{m1} = 3.316 \text{ mS}$, $g_{m2} = 3.162 \text{ mS}$, $r_{ds1} = 50 \text{ k}\Omega$ and $r_{ds2} = 40 \text{ k}\Omega$.

$$\therefore R_{out} = \frac{50 \text{ k}\Omega \| 40 \text{ k}\Omega}{1 + 0.5 \left(\frac{3316 + 3162}{25 + 20} \right)} = \frac{22.22 \text{ k}\Omega}{1 + 0.5(143.9)} = 304 \Omega \quad R_{out} = 5.42 \text{ k}\Omega \text{ if } R_L = 1 \text{ k}\Omega$$

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Summary: Multistage Amplifiers

- Cascodes
- Cascades
- Folded cascodes
- Output stages
 - Source follower
 - Push-pull source follower
 - Push-pull common-source
 - Negative shunt feedback



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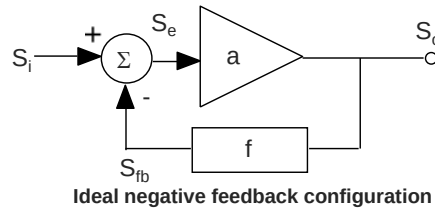
Introduction: Feedback

- General feedback structure
- Effects of negative feedback
 - Gain sensitivity
 - Nonlinear distortion
 - Frequency response
- Feedback topologies – shunt-shunt, series-series, series-shunt, shunt-series
 - Identification of topology
 - Amplifier configuration
 - Derivation of open-loop gain, feedback factor, closed-loop gain, input and output impedances
 - Examples and solutions



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Feedback Concepts: Ideal Feedback Equation



Network

$$S_e = S_i - S_{fb} = S_i - fS_o \quad S_o = aS_e = aS_i - afS_o$$

$$A = \frac{S_o}{S_i} = \frac{a}{1+af} \Rightarrow \text{closed loop gain } T = af \rightarrow \text{loop gain}$$

$$A = \frac{S_o}{S_i} = \frac{a}{1+T} \text{ for } T \gg 1, A \rightarrow \frac{1}{f}$$

$$\text{Feedback signal/input signal} = \frac{S_{fb}}{S_i} = \frac{T}{1+T}$$

$$\text{Error signal/input signal} = \frac{S_e}{S_i} = \frac{1}{1+T}$$



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Feedback Concepts: What Do Gains Indicate?

- Need to be careful to distinguish among:
 - Open-loop gain A
 - Closed-loop gain $\frac{A}{1 + A\beta}$
 - Loop gain $A\beta$
- Loop gain $A\beta$ is *very important* in feedback systems!
 - Tells you frequency range for which feedback is working $|A\beta| > 1$
 - Stability criterion: Check phase of $A\beta$ at frequency for which $|A\beta| = 1$



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Feedback Concepts: Gain Sensitivity

Sensitivity of the closed-loop gain with respect to the open-loop gain:

$$\frac{dA}{da} = \frac{1+af-af}{(1+af)^2} = \frac{1}{(1+T)^2} = \frac{1}{a} \frac{a}{(1+T)^2} = \frac{A}{a} \frac{1}{1+T}$$

$$\frac{dA}{A} = \frac{1}{(1+T)} \frac{da}{a}$$

Example

An amplifier with a gain of 10 has a gain variation of 3.3%. Use these amplifiers to design an amplifier with a gain of 10 and a gain variation < 0.1%.

Solution

You must cascade several amplifiers together and feedback around the cascade with feedback $dA/A = 0.001$. The following is for three cascaded stages:

$$a = 1000 \text{ and } \frac{da}{a} = \frac{[10(1+0.333)^3] - 1000}{1000} = 0.1023$$

$$\frac{dA}{A} = \frac{1}{(1+T)} \frac{da}{a} = \frac{0.1023}{1+T} = 0.001 \Rightarrow T = 101$$

$$A = \frac{1000}{1+101} = 9.80 \quad a = 1000 \quad f = \frac{101}{1000} = 0.101$$

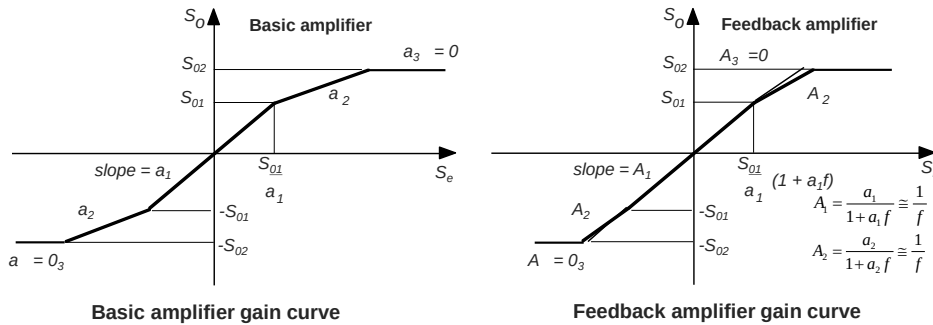
Gain is close to 10 but not exactly; however, the tolerance is 0.1%.



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Nonlinear Distortion: Negative-Feedback Effects

The major cause of nonlinear distortion in amplifiers is due to changes of gain with the signal level.



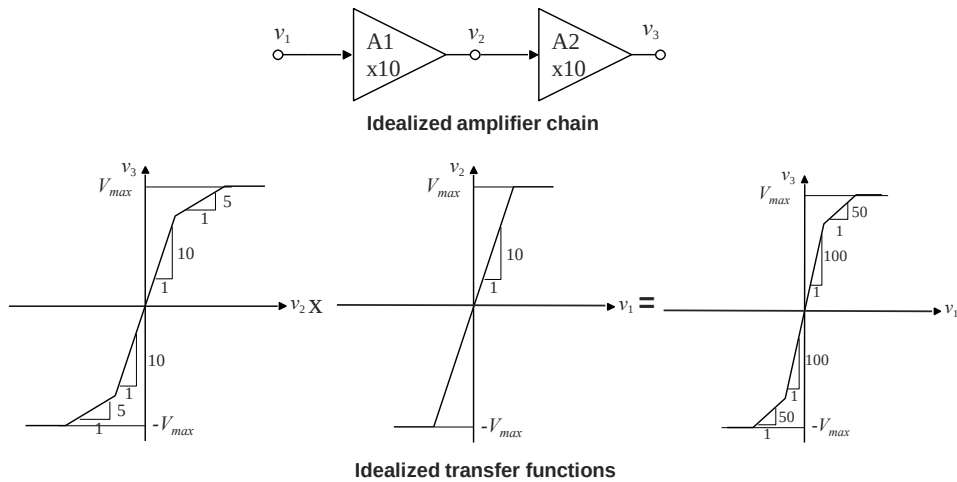
- Three gain regions exist for each case.
- Horizontal scale compression for feedback amplifier.
- No distortion improvement for hard saturation cases.
- Distortion improvement with feedback $\approx (1 + T)$.



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Demonstration: Reduction of Nonlinearity

You can reduce nonlinearity with negative feedback. For this demonstration consider the following circuit and assume that the amplifiers have the following voltage-transfer functions:



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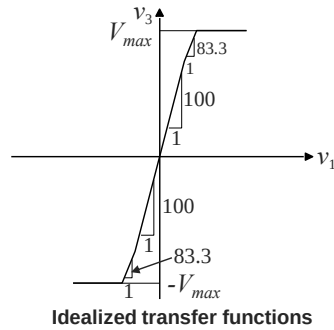
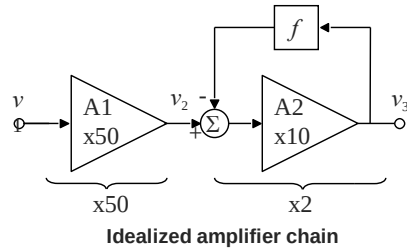
Demonstration (continued)

You can apply feedback around the second stage, which results in a gain of x2 and increase the gain of the first stage from x10 to x50.

To find the value of f , you can solve the following:

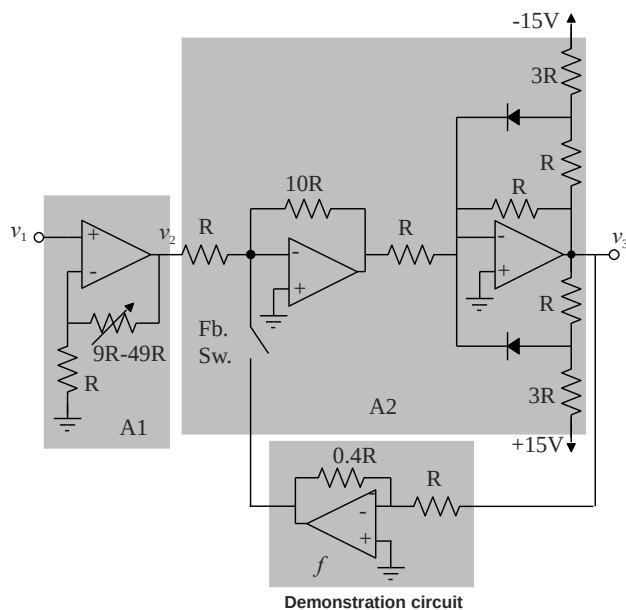
$$\frac{v_3}{v_2} = \frac{A_2}{1 + A_2 f} = 2 \Rightarrow f = 0.4$$

The resulting transfer function is shown on the graph.



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Demonstration (continued)



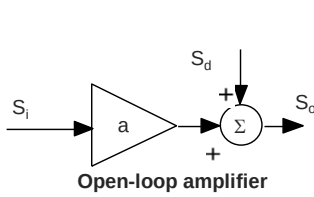
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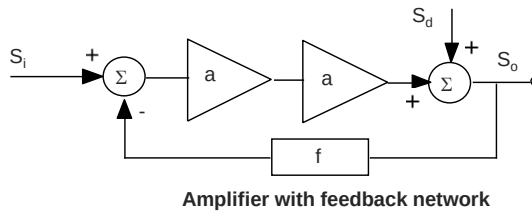
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Negative-Feedback Effects on Signal Distortion

Consider the output stage of an amplifier where distortion occurs because this stage is being driven very hard and a distortion signal S_d is introduced. Examine the two cases – with and without feedback.



$$S_o = aS_i + S_d$$



$$S_o = \frac{a^2 S_i}{1 + a^2 f} + \frac{S_d}{1 + a^2 f}$$

Choose: $f = \frac{a-1}{a^2} \rightarrow s_o = aS_i + \frac{S_d}{a}$

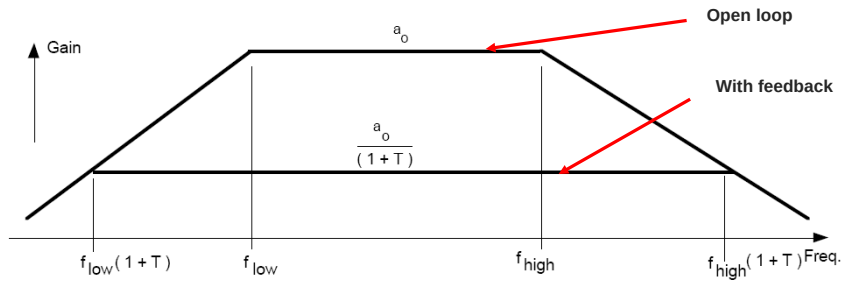
- Desired output signal is the same in both cases.
- With feedback, distortion is reduced by gain a .
- $T = a2f = a - 1$
- Improved performance with low-gain amplifier.



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Negative Feedback Effects (continued)

When negative feedback is used with an amplifier having a single dominant low-frequency and a single dominant high-frequency pole, gain, and bandwidth can be traded evenly as shown.



Idealized frequency response with and without feedback

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Different Types of Opamp Nonidealities

		Static (DC)	Dynamic (AC)
Linear	Error	Offset voltage DC bias current	Finite bandwidth
	Text	Section 2.7	Section 2.5
	Analysis Strategy	Add error sources external to ideal opamp; total effects using superposition	Determine feedback factor b with all inputs = 0; closed loop $f_{3dB} = b f_T$
Nonlinear	Error	Output voltage swing limit Output current limit	Slew rate limit
	Text	Sec. 2.6.1, 2.6.2	Sec. 2.6.3
	Analysis Strategy	Determine maximum total v_{OUT} , i_{OUT} , at opamp output from linear system model; compare to limits	Determine maximum $dv_{OUT}(t)/dt$ from linear system behavior (transfer function for sine wave, general exponential response for step); compare to limit

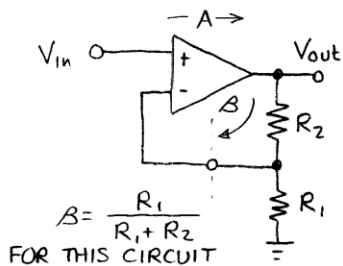
Reference: http://ece.wpi.edu/~mcneill/handouts/Op-Amp_Error_Matrix_S&S.html



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Opamp as Classical Feedback System

OP-AMP CIRCUIT:



β : "FEEDBACK FACTOR" FRACTION OF V_{out} FED BACK TO V_-

FIND CLOSED LOOP GAIN $G = \frac{V_{out}}{V_{in}}$

$$V_{out} = A(V_{in} - \beta V_{out}) \Rightarrow \boxed{\frac{V_{out}}{V_{in}} = \frac{A}{1 + A\beta} = G}$$

✓ AS EXPECTED
 $G \rightarrow \frac{1}{\beta}$
AS $A \rightarrow \infty$

Reference: <http://ece.wpi.edu/~mcneill/handouts/opampfeedbackgain-bw-prod.pdf>

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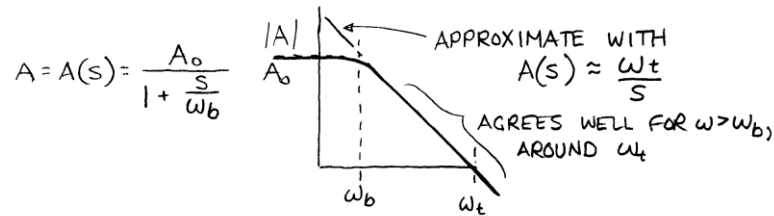


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Closed-Loop Gain When Open-Loop Gain Is Finite

What is the closed-loop gain when the open-loop gain is finite?

REAL OP-AMP HAS SINGLE POLE TRANSFER FUNCTION



SUBSTITUTE IN G EXPRESSION

$$G = \frac{A}{1 + A\beta} = \frac{\frac{\omega_t}{s}}{1 + \beta \frac{\omega_t}{s}} \quad \left. \begin{array}{l} \text{MULTIPLY NUM, DEN} \\ \text{BY } \frac{s}{\beta \omega_t} \end{array} \right\}$$

MESSAGE INTO GENERAL FIRST ORDER FORM

$$G = \frac{[\text{DC GAIN}]}{1 + \frac{s}{[\omega_{3dB}]}}$$



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First-Order Transfer Function: DC Gain, f3 dB Bandwidth

$$G = \frac{\frac{\omega_t}{s} \frac{s}{\beta \omega_t}}{\left(1 + \beta \frac{\omega_t}{s}\right) \frac{s}{\beta \omega_t}} \Rightarrow \boxed{\frac{\frac{1}{\beta}}{1 + \frac{s}{\beta \omega_t}}}$$

$\left. \frac{1}{\beta} \right\} \text{DC GAIN}$
 $\left. 1 + \frac{s}{\beta \omega_t} \right\} \text{CLOSED LOOP BANDWIDTH } \omega_{3dB}$

GAIN-BANDWIDTH TRADEOFF!:

WHEN WE CHOOSE A FEEDBACK FACTOR β (FRACTION < 1)

TO GET A CLOSED LOOP DC GAIN OF $\frac{1}{\beta}$

CLOSED LOOP BANDWIDTH ω_{3dB} IS $\beta \omega_t$

UNITY GAIN FREQUENCY ω_t REDUCED BY β FACTOR

INVERSE RELATIONSHIP: PRODUCT IS CONSTANT

$$\left[\begin{array}{c} \text{CLOSED LOOP} \\ \text{GAIN} \end{array} \right] \times \left[\begin{array}{c} \text{CLOSED LOOP} \\ \text{BANDWIDTH} \end{array} \right] = \left[\begin{array}{c} \text{GAIN-BANDWIDTH} \\ \text{PRODUCT} \end{array} \right]$$



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Gain-Bandwidth: Voltage Feedback Opamps

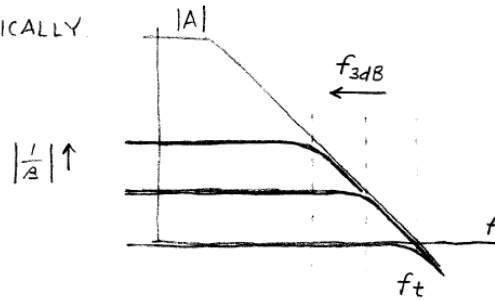
UNITY GAIN FREQUENCY ω_t REDUCED BY β FACTOR

INVERSE RELATIONSHIP: PRODUCT IS CONSTANT

$$\left[\begin{array}{c} \text{CLOSED LOOP} \\ \text{GAIN} \end{array} \right] \times \left[\begin{array}{c} \text{CLOSED LOOP} \\ \text{BANDWIDTH} \end{array} \right] = \left[\begin{array}{c} \text{GAIN-BANDWIDTH} \\ \text{PRODUCT} \end{array} \right]$$

$\frac{1}{\beta}$ $\underbrace{\beta \omega_t}_{\omega_{3dB}}$ $\underbrace{\omega_t}_{\text{UNITY GAIN FREQUENCY}}$

GRAPHICALLY



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IC Design vs. Discrete Design

- Opamp design: gain-bandwidth product f_T (unity gain frequency) is determined by application requirements for:
 - Closed-loop gain
 - Closed-loop bandwidth f_{3dB}
- Discrete design problem
 - Select opamp to meet f_T required for application
- Integrated design problem
 - *Design* opamp which requires f_T for application



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Opamp as Integrator

- Approximate as integrator:
 ω_T is unity gain frequency

$$\frac{v_{out}}{v_{id}} = \frac{1}{s \tau_T} \quad \tau_T = \frac{1}{\omega_T}$$

- In time domain:

$$v_{out} = \frac{1}{\tau_T} \int v_{id} dt$$

- Take d/dt of both sides:

$$\frac{d}{dt} v_{out} = \frac{v_{id}}{\tau_T} = \frac{v_+ - v_-}{\tau_T}$$

- Result:

- If $v_- \neq v_+$, output v_{out} changes.
- Rate of change is proportional to size of differential input.
- Negative feedback: Circuit is configured so change reduces $v_+ - v_-$ difference.
- Equilibrium ($d/dt=0$) when $v_- = v_+$.



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Opamp as Integrator (continued)

- Negative Feedback: $v_- = v_+$.
- How can amplifier have nonzero output for zero input?
- Integrator: zero steady state error.
- Output at equilibrium (any value!) for zero input.
- Only need (integrated) past history to sustain nonzero output.



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Formal Treatment of Feedback

- Four basic amplifiers
- Four basic feedback configurations
 - Shunt-shunt
 - Series-series
 - Series-shunt
 - Shunt-series
- Identification procedure
- Examples



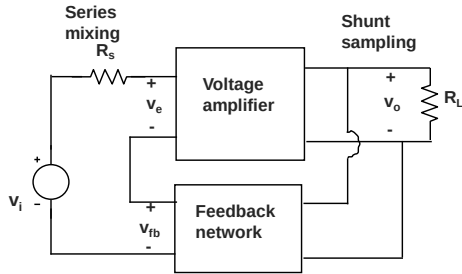
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Four Basic Amplifiers

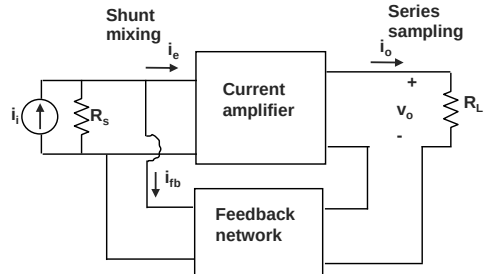
Parameter	Amplifier type			
	Voltage	Transconductance	Transresistance	Current
Small-signal model				
Ideal forward gain	$v_o = A_v v_e$	$i_o = G_m v_e$	$v_o = R_m i_e$	$i_o = A_i i_e$
Ideal R_i	$\rightarrow \infty \Omega$	$\rightarrow \infty \Omega$	$\rightarrow 0 \Omega$	$\rightarrow 0 \Omega$
Ideal R_o	$\rightarrow 0 \Omega$	$\rightarrow \infty \Omega$	$\rightarrow 0 \Omega$	$\rightarrow \infty \Omega$
S_i, S_{fb} and S_e	Voltage	Voltage	Current	Current
S_o variable	Voltage	Current	Voltage	Current
SS model with source and load				
Ideal source, R_S	$R_S = 0$ or $R_S \ll R_i$	$R_S = 0$ or $R_S \ll R_i$	$R_S = \infty$ or $R_S \gg R_i$	$R_S = \infty$ or $R_S \gg R_i$
Ideal load, R_L	$R_L = \infty$ or $R_L \gg R_o$	$R_L = 0$ or $R_L \ll R_o$	$R_L = \infty$ or $R_L \gg R_o$	$R_L = 0$ or $R_L \ll R_o$
Overall forward gain	$\frac{R_i R_L A_v}{(R_S + R_i)(R_L + R_o)}$	$\frac{R_i R_o G_m}{(R_S + R_i)(R_L + R_o)}$	$\frac{R_S R_L R_m}{(R_S + R_i)(R_L + R_o)}$	$\frac{R_S R_L A_i}{(R_S + R_i)(R_L + R_o)}$

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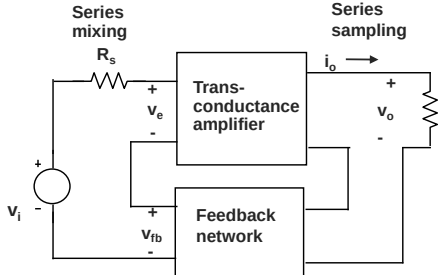
Four Basic Feedback Topologies



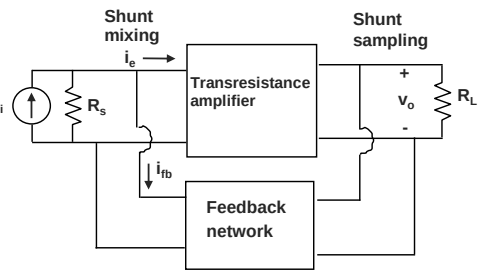
Series-shunt feedback topology



Shunt-series feedback topology



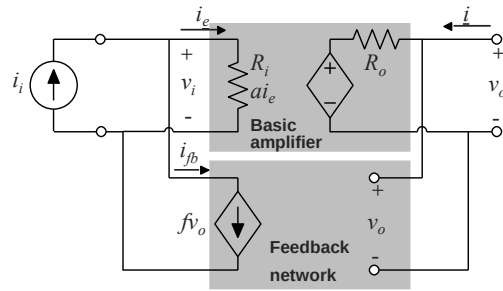
Series-series feedback topology



Shunt-shunt feedback topology

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Shunt-Shunt Feedback Amplifiers



Shunt-shunt configuration

$$v_o = ai_e = a(i_i - i_{fb}) = ai_i - ai_{fb} = av_i - afv_o \rightarrow v_o(1 + af) = ai_i \rightarrow \frac{i_o}{i_i} = \frac{a}{1 + af}$$

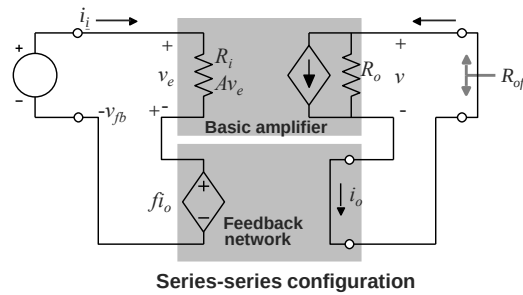
$$R_{if} = \frac{v_i}{i_i} = \frac{i_e R_i}{i_e + i_{fb}} = \frac{i_e R_i}{i_e + fi_o} = \frac{i_e R_i}{i_e + afi_e} = \frac{R_i}{1 + af} = \frac{R_i}{1 + T}$$

$$R_{of} = \left. \frac{v_o}{i} \right|_{i_i=0} \quad v_o = R_o i + ai_e = R_o i + a(-i_{fb}) = R_o i - afv_o \rightarrow v_o(1 + af) = R_o i$$

$$\therefore R_{of} = \frac{v_o}{i} = \frac{R_o}{1 + af} = \frac{R_o}{1 + T}$$

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Series-Series Feedback Amplifiers



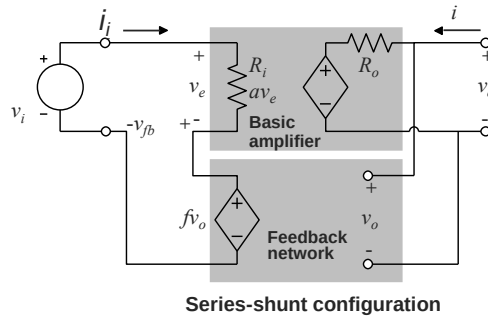
$$i_o = av_e = a(v_i - v_{fb}) = av_i - av_{fb} = av_i - afi_o \rightarrow i_o(1 + af) = av_i \rightarrow \frac{i_o}{v_i} = \frac{a}{1 + af}$$

$$R_{if} = \frac{v_i}{i_i} = \frac{v_i}{v_e} R_i = R_i \frac{v_e + v_{fb}}{v_e} = R_i \frac{v_e + f i_o}{v_e} = R_i \frac{v_e + af v_e}{v_e} = R_i(1 + af) = R_i(1 + T)$$

$$R_{of} = \left. \frac{v_o}{i_o} \right|_{i_i=0} \quad R_{of} = \frac{v}{i_o} = \frac{R_o(i_o - av_e)}{i_o} = \frac{R_o(i_o + af i_o)}{i_o} = R_o(1 + af)$$

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Series-Shunt Feedback Amplifiers



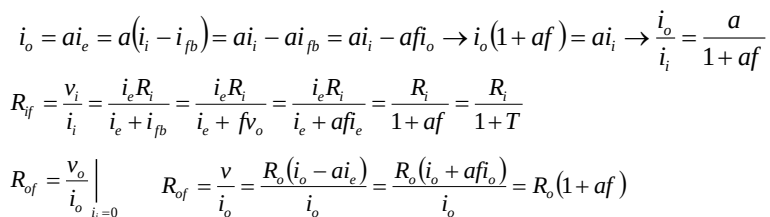
Find the equivalent voltage gain, input resistance (R_{if}), and output resistance (R_{of}):

$$v_o = a v_e = a(v_i - v_{fb}) = a v_i - a v_{fb} = a v_i - a f v_o \rightarrow v_o(1 + a f) = a v_i \rightarrow \frac{v_o}{v_i} = \frac{1}{1 + a f}$$

$$R_{if} = \frac{v_i}{i_i} = \frac{v_i}{v_e} R_i = R_i \frac{v_e + v_{fb}}{v_e} = R_i \frac{v_e + f v_o}{v_e} = R_i \frac{v_e + a f v_e}{v_e} = R_i(1 + a f) = R_i(1 + T)$$

$$R_{of} = \frac{v_o}{i} \bigg|_{v_i=0} = \frac{v_o - a v_e}{R_o} = \frac{v_o - a(-v_{fb})}{R_o} = \frac{v_o - a f v_o}{R_o} \Rightarrow R_{of} = \frac{R_o}{1 + a f} = \frac{R_o}{1 + T}$$

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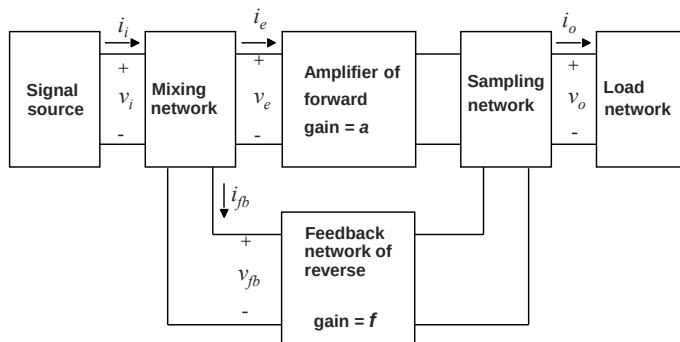
Feedback Topologies: Summary

Parameter	Amplifier Type			
	Voltage	Transconductance	Transresistance	Current
Feedback Topology	Series-shunt	Series-series	Shunt-shunt	Shunt-series
S_i , S_{fb} and S_e	Voltage	Voltage	Current	Current
S_o Variable	Voltage	Current	Voltage	Current
Closed Loop Gain	$A_{vf} = \frac{A_v}{1+fA_v}$	$G_{mf} = \frac{G_m}{1+fG_m}$	$R_{mf} = \frac{R_m}{1+fR_m}$	$A_{if} = \frac{A_i}{1+fA_i}$
Closed Loop Input Resistance	$R_{if} = R_i(1+fA_v)$	$R_{if} = R_i(1+fG_m)$	$R_{if} = \frac{R_i}{1+fR_m}$	$R_{if} = \frac{R_i}{1+fA_i}$
Closed Loop Output Resistance	$R_{of} = \frac{R_o}{1+fA_v}$	$R_{of} = R_o(1+fG_m)$	$R_{of} = \frac{R_o}{1+fR_m}$	$R_{of} = R_o(1+fA_i)$



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Two-Terminal Representation

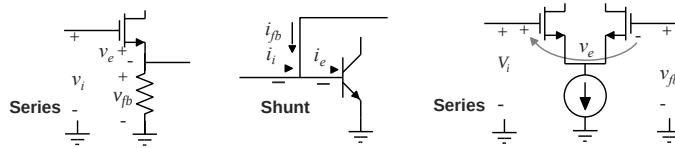


Two-terminal representation of single-loop, negative feedback system

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Feedback Topology Identification Procedure

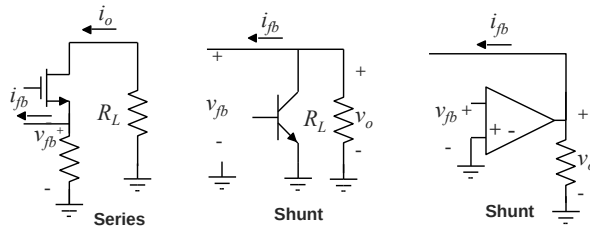
1. Identify the feedback loop by tracing around the feedforward and feedback path. Also check to see if the feedback is positive or negative.
2. Identify whether the mixing network is series or shunt. If the signal source has one terminal on AC ground, then:
 - If the input active device has one of its input terminals on AC ground, then the mixing network must be shunt.
 - If the signal and feedback sources are applied to different input terminals of the input active device, then the mixing network is series. (This includes differential amplifiers where two devices form the input active device.)
 - If the signal source does not have one of its input terminals on AC ground, or to check the previous steps, try assigning the variables x_i , x_{fb} , and x_e on the schematic in such a manner as to implement the equation $x_e = x_i \pm x_{fb}$. If this equation can be written using voltages (currents), then the mixing circuit is series (shunt).



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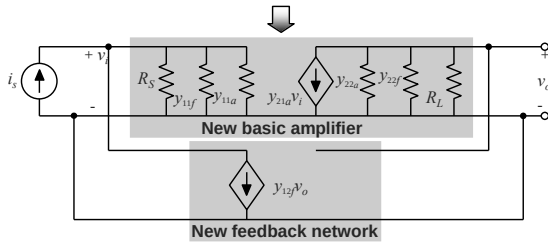
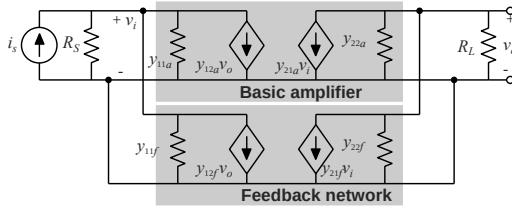
Feedback Topology Identification Procedure (continued)

3. Identify the sampling circuit as series or shunt. If the load is grounded, then
 - If the output active device has one of its two possible output terminals grounded, then the feedback is shunt.
 - If the output active device has neither of its output terminals on ground and if the output signal is taken from one of its output terminals and the feedback signal from the other output terminal, then the feedback is series.
 - If the load is not grounded, or to check the above test, identify the load resistor R_L and apply the following test:
 - If x_{fb} becomes zero when $R_L = 0$, then the sampling network is shunt.
 - If x_{fb} becomes zero when $R_L = \infty$, then the sampling network is series.

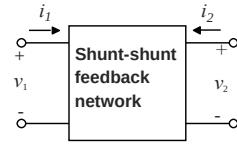


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Shunt-Shunt Feedback



Shunt-shunt configuration



Signal conventions

$$i_1 = y_{11}v_1 + y_{12}v_2$$

$$i_2 = y_{21}v_1 + y_{22}v_2$$

where for the new basic amplifier

$$y_{11} = \left. \frac{i_1}{v_1} \right|_{v_2=0} = G_S + y_{11a} + y_{11f}$$

$$y_{12} = \left. \frac{i_1}{v_2} \right|_{v_1=0} = 0 \quad y_{21} = \left. \frac{i_2}{v_1} \right|_{v_2=0} = y_{21a}$$

$$y_{22} = \left. \frac{i_2}{v_2} \right|_{v_1=0} = G_L + y_{22a} + y_{22f}$$

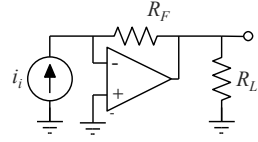
$$\frac{v_o}{i_s} = \frac{v_2}{i_1} = A = \left(\frac{a}{1+af} \right) = \frac{(-y_{21a}/y_{11}y_{22})}{1+(-y_{21a}/y_{11}y_{22})y_{12f}} \Rightarrow a = \frac{-y_{21a}}{y_{11}y_{22}} \text{ and } f = y_{12f}$$



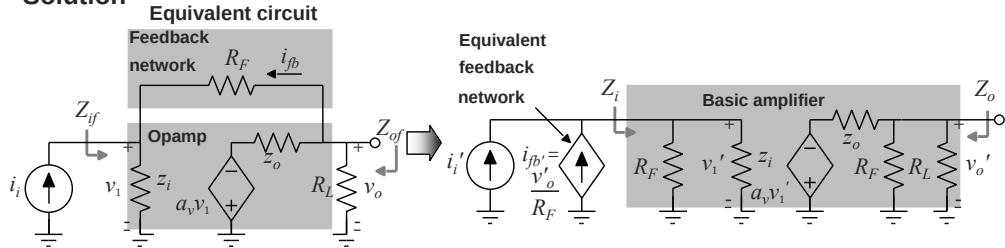
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Example: Inverting Opamps

Find the closed-loop transfer function, A , the closed-loop input resistance, Z_{if} , and the closed-loop output resistance, Z_{of} of the shunt-shunt configuration shown. The opamp has a differential input resistance of z_i , voltage gain of a_v , and output resistance of z_o . In the diagram below, primed variables indicate an open loop.



Solution



It can be shown that $f = y_{12f} = \frac{i_1'}{v_2'} \bigg|_{v_1'=0} = \frac{-1}{R_f}$

The forward gain a is

$$a = \frac{v_o'}{i_i'} = \left(\frac{v_o'}{v_1'} \right) \left(\frac{v_1'}{i_i'} \right) = \left(\frac{-a_v (R_F \parallel R_L)}{z_o + (R_F \parallel R_L)} \right) \left(\frac{z_i R_F}{z_i + R_F} \right) = \left(\frac{-a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i R_F}{z_i + R_F} \right) \text{ where } i_{fb}' = 0$$

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Example: Inverting Opamps (continued)

The loop gain is:

$$T = af = \left(\frac{a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i}{z_i + R_F} \right)$$

The closed-loop gain is:

$$\frac{v_o}{i_i} = \frac{a}{1 + af} = \frac{\left(\frac{-a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i R_F}{z_i + R_F} \right)}{1 + \left(\frac{a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i}{z_i + R_F} \right)} = \frac{(-a_v R_F^2 R_L z_i)}{(z_o R_L + z_o R_L + R_F R_L)(z_i + R_F) + a_v R_F R_L z_i}$$

The closed-loop input impedance is:

$$Z_{if} = \frac{Z_o}{1 + T} = \frac{\frac{z_i R_F}{z_i + R_F}}{1 + \left(\frac{a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i}{z_i + R_F} \right)} \approx \frac{z_o R_L + z_o R_L + R_F R_L}{a_v R_L}$$

The closed-loop output impedance is:

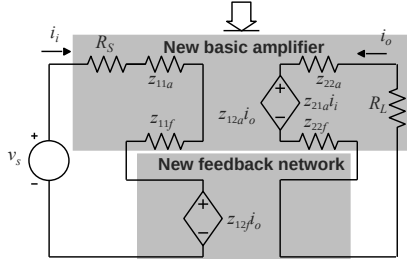
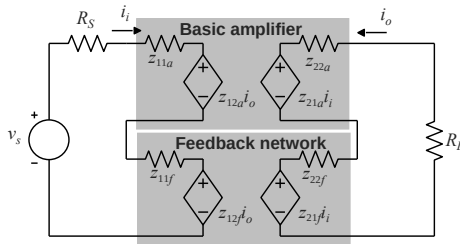
$$Z_{of} = \frac{Z_o}{1 + T} = \frac{\frac{z_o \| R_F \| R_L}{1 + \left(\frac{a_v R_F R_L}{z_o R_L + z_o R_L + R_F R_L} \right) \left(\frac{z_i}{z_i + R_F} \right)}}$$

If $a_v = 200,000$, $z_i = 2 \text{ M}\Omega$, and $z_o = 75\Omega$, $R_F = 1 \text{ M}\Omega$, and $R_L = 10 \text{ k}\Omega$, then $T = 133,333$, $Z_i = 2 \| 1 = 0.667 \text{ M}\Omega \rightarrow Z_{if} = 5\text{W}$, $Z_o \gg 75\Omega \rightarrow Z_{of} = 0.563 \text{ m}\Omega$ and $A = -999,992\Omega$

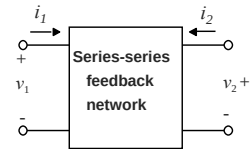


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Series-Series Feedback



Series-series configuration



Signal conventions

$$v_1 = z_{11}i_1 + z_{12}i_2$$

$$v_2 = z_{21}i_1 + z_{22}i_2$$

Where for the new basic amplifier

$$z_{11} = \left. \frac{v_1}{i_1} \right|_{i_2=0} = R_s + z_{11a} + z_{11f}$$

$$z_{12} = \left. \frac{v_1}{i_2} \right|_{i_1=0} = 0 \quad z_{21} = \left. \frac{v_2}{i_1} \right|_{i_2=0} = z_{21a}$$

$$z_{22} = \left. \frac{v_2}{i_2} \right|_{i_1=0} = R_L + z_{22a} + z_{22f}$$

$$\frac{i_o}{v_s} = \frac{i_2}{v_1} = A = \frac{a}{1+af} = \frac{(-z_{21a}/z_{11}z_{22})}{1+(-z_{21a}/z_{11}z_{22})z_{12f}} \Rightarrow a = \frac{-z_{21a}}{z_{11}z_{22}} \text{ and } f = z_{12f}$$

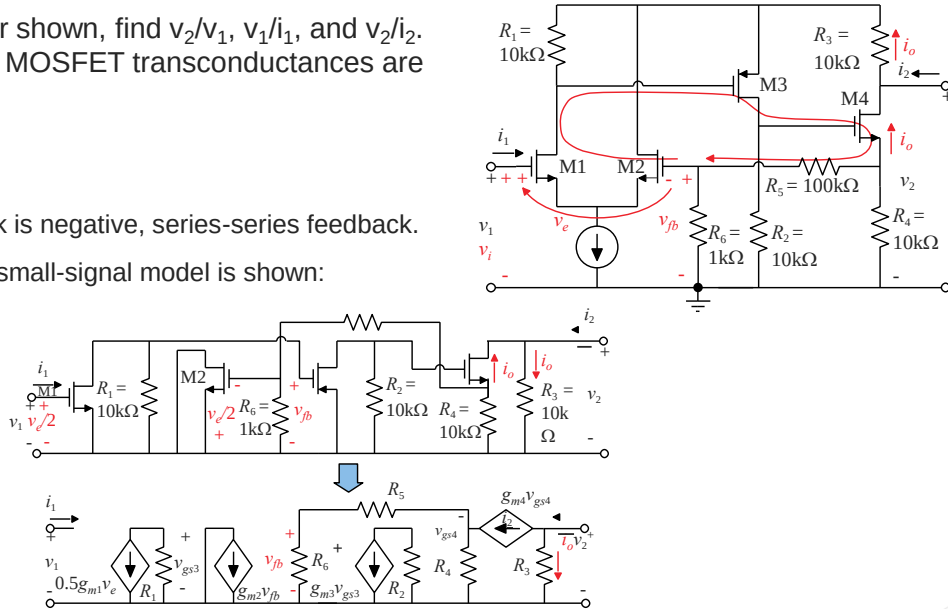
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Example: Series-Series Feedback Triple

For the amplifier shown, find v_2/v_1 , v_1/i_1 , and v_2/i_2 . Assume that all MOSFET transconductances are 1 mS.

Solution

1. The feedback is negative, series-series feedback.
2. Closed-loop small-signal model is shown:



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Example: Series-Series Feedback Triple (continued)

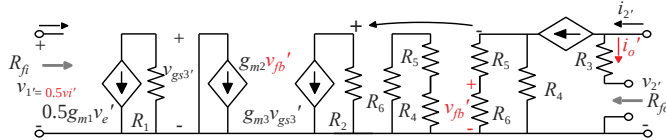
3. Break open the loop by finding the loading effects of the feedback network on the amplifier. Find z_{11f} and z_{22f} :

$$z_{11f} = \left. \frac{v_{1f}}{i_{1f}} \right|_{i_{2f}=i_o=0} = R_4 + R_5 \quad z_{22f} = \left. \frac{v_{2f}}{i_{2f}} \right|_{i_{1f}=i_i=0} = R_5 + R_6$$

3. The open-loop model is:

$$f = \frac{v_{fb}'}{i_o'} = \frac{-R_4 R_6}{R_4 + R_5 + R_6}$$

$$= \frac{-10 \cdot 1}{10 + 100 + 1} k\Omega = -909 \Omega$$



$$a = \frac{i_o'}{v_i'} = \left(\frac{i_o'}{v_{gs4}'} \right) \left(\frac{v_{gs4}'}{v_{gs3}'} \right) \left(\frac{v_{gs3}'}{v_i'} \right) = (-g_{m4}) \left(\frac{-g_{m3}R_2}{1 + g_{m4}[R_4 \parallel (R_5 + R_6)]} \right) - \frac{gm_1 R_1}{2}$$

$$= (-1mS)(-0.99)(-5) = -4.95mS \Rightarrow T = af = 4.5$$

4. Input resistance, v_1/i_1 :

$$R_i = \infty \rightarrow R_{if} = R_i(1 + T) = 5.5\infty = \infty$$

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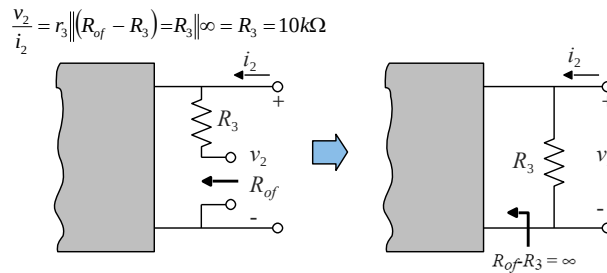
Example: Series-Series Feedback Triple (continued)

Voltage gain, v_2/v_1 . First find i_o/v_i :

$$\frac{i_o}{v_i} = \frac{a}{1+af} = \frac{-4.95mS}{1+4.5} = -0.9mS \quad (1/f = -1.1mS)$$

$$\frac{v_2}{v_1} = \left(\frac{i_o R_3}{v_i} \right) = (-0.9mS)(10k\Omega) = -9.0V/V$$

Output resistance is v_2/i_2 , however, because you have assumed that $r_{ds4} = \infty$, then:



(R_3 is not really inside the feedback loop and therefore is not influenced by the feedback.)

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Summary: Series-Series Feedback

Series-series feedback increases both the input and output impedances. If the mixing circuit is associated with the collector or drain, then the output resistance designated as v_2/i_2 is found by finding R_{of} (which is the resistance in series with the output feedback loop) and using the following relationship:

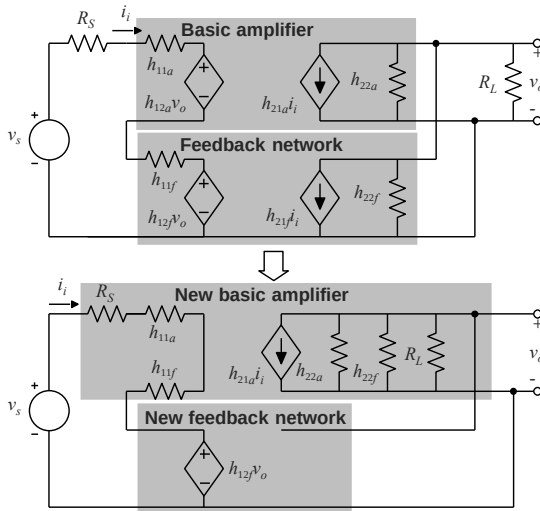
$$\frac{v_2}{i_2} = R_L \parallel R_{of} - R_L$$

- If the mixing circuit is associated with the emitter or source, then the output resistance designated as v_2/i_2 is equal to R_L .
- If $T \gg 1$, then $A \approx 1/f$. The product of a and f must always be *positive* for negative feedback.

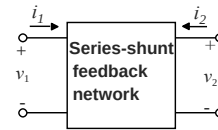


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Series-Shunt Feedback



Series-shunt configuration



Signal conventions

$$v_1 = h_{11}i_1 + h_{12}v_2$$

$$i_2 = h_{21}i_1 + h_{22}v_2$$

Where for the new basic amplifier:

$$h_{11} = \left. \frac{v_1}{i_1} \right|_{v_2=0} = R_S + h_{11a} + h_{11f}$$

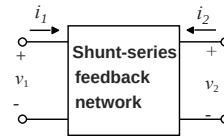
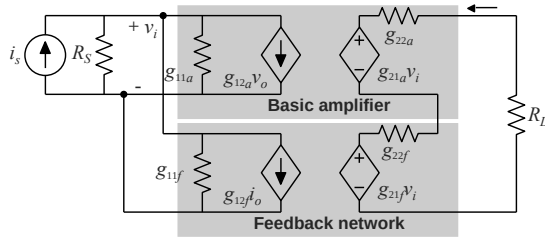
$$h_{12} = \left. \frac{v_1}{v_2} \right|_{i_1=0} = 0 \quad h_{21} = \left. \frac{i_2}{i_1} \right|_{v_2=0} = h_{21a}$$

$$h_{22} = \left. \frac{i_2}{v_2} \right|_{i_1=0} = G_L + h_{22a} + h_{22f}$$

$$\frac{v_o}{v_s} = \frac{v_2}{v_1} = A = \frac{a}{1+af} = \frac{(-h_{21a}/h_{11}h_{22})}{1+(-h_{21a}/h_{11}h_{22})h_{12f}} \Rightarrow a = \frac{-h_{21a}}{h_{11}h_{22}} \text{ and } f = h_{12f}$$

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Shunt-Series Feedback



Signal conventions

$$i_1 = g_{11}v_1 + g_{12}i_2$$

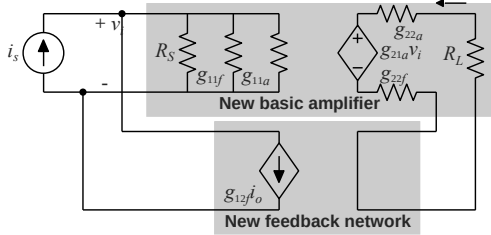
$$v_2 = g_{21}v_1 + g_{22}i_2$$

Where for the new basic amplifier:

$$g_{11} = \left. \frac{i_1}{v_1} \right|_{i_2=0} = G_S + g_{11a} + g_{11f}$$

$$g_{12} = \left. \frac{v_2}{v_1} \right|_{i_2=0} = g_{21a} \quad g_{12} = \left. \frac{i_1}{i_2} \right|_{v_1=0} = 0$$

$$g_{22} = \left. \frac{v_2}{i_2} \right|_{v_1=0} = R_L + g_{22a} + g_{22f}$$



Shunt-series configuration

$$\frac{i_o}{i_s} = \frac{i_2}{i_1} = A = \frac{a}{1+af} = \frac{(-g_{21a}/g_{11}g_{22})}{1+(-g_{21a}/g_{11}g_{22})g_{12f}} \Rightarrow a = \frac{-g_{21a}}{g_{11}g_{12}} \text{ and } f = g_{12f}$$



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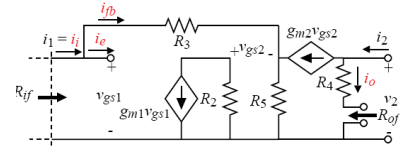
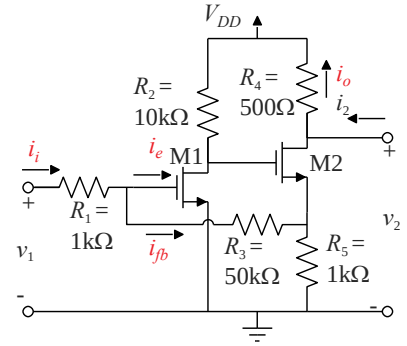
Example: Shunt-Series Feedback Amplifiers

For the amplifier shown, find v_2/v_1 , v_1/i_1 , and v_2/i_2 . Assume that all MOSFET transconductances are 1 mS.

Solution

1. Identify topology: You can see that the circuit is shunt-series, negative feedback. Also, notice that R_1 is outside the feedback circuit.
2. Break open the loop (closed-loop small-signal model is shown in the following figure on the lower right) by finding the loading effects of the feedback network on the amplifier, which involves finding h_{11f} and h_{22f} .

$$g_{11f} = \left. \frac{i_{1f}}{v_{1f}} \right|_{i_2=i_o=0} = 1/(R_1 + R_3) \text{ and } g_{22f} = \left. \frac{v_{2f}}{i_{2f}} \right|_{v_1=v_i=0} = R_3 \parallel R_5$$



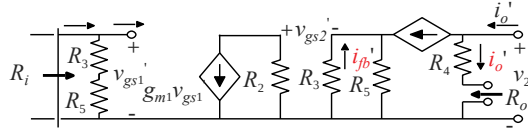
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Example: Shunt-Series Feedback Amplifiers (continued)

3. Open-circuit small-signal model:

$$f = \frac{i_{fb}'}{i_o'} = \frac{R_5}{R_3 + R_5}$$

$$= \frac{1}{1 + 50} = \frac{1}{51} = 0.0196$$



$$a = \frac{i_o'}{i_i'} = \left(\frac{i_o'}{v_{gs2}'} \right) \left(\frac{v_{gs2}'}{v_{gs1}'} \right) \left(\frac{v_{gs1}'}{i_i'} \right) = (-g_{m2}) \left(\frac{-g_{m1}R_2}{1 + g_{m2}(R_3 \parallel R_5)} \right) (R_3 + R_5)$$

$$= (-1 \times 10^{-3} \text{ S}) (-5.0495) (51 \times 10^3 \Omega) = 257.5 \text{ A/A} \Rightarrow T = \frac{257.5}{51} = 5.05$$

4. Input resistance,
- v_1/i_1
- :

$$R_i = R_3 + R_5 = 51 \text{ k}\Omega \rightarrow R_{if} = \frac{R_i}{1 + T} = \frac{51 \text{ k}\Omega}{6.05} = 8.43 \text{ k}\Omega$$

5. However,
- v_1/i_1
- , can be found as:

$$v_1/i_1 = R_{in} = R_s + R_{if} = 1 \text{ k}\Omega + 8.43 \text{ k}\Omega = 9.43 \text{ k}\Omega$$

6. Voltage gain,
- v_2/v_1
- . First find
- i_o/i_i

$$\frac{i_o}{i_i} = \frac{a}{1 + af} = \frac{257.5}{1 + 5.05} = 42.56 \text{ A/A} \rightarrow \frac{v_2}{v_1} = \left(\frac{i_o}{i_i} \right) \left(\frac{R_4}{R_{in}} \right) = (42.56) \left(\frac{500 \Omega}{9.43 \text{ k}\Omega} \right) = \underline{\underline{2.257 \text{ V/V}}}$$



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Example: Shunt-Series Feedback Amplifiers (continued)

7. Output resistance, v_2/i_2 . First find R_o .

$R_o = \infty$ because it includes the drain of M2 and $r_{ds2} = \infty$.

$$\therefore \frac{v_2}{i_2} = R_4 \parallel R_{of} - R_4 = 500\Omega \parallel \infty = \underline{\underline{500\Omega}}$$

What about i_2/i_1 ? Assume the output is shorted, then:

$$i_2/i_1 = -i_o/i_i = -42.56 \text{ A/A}$$



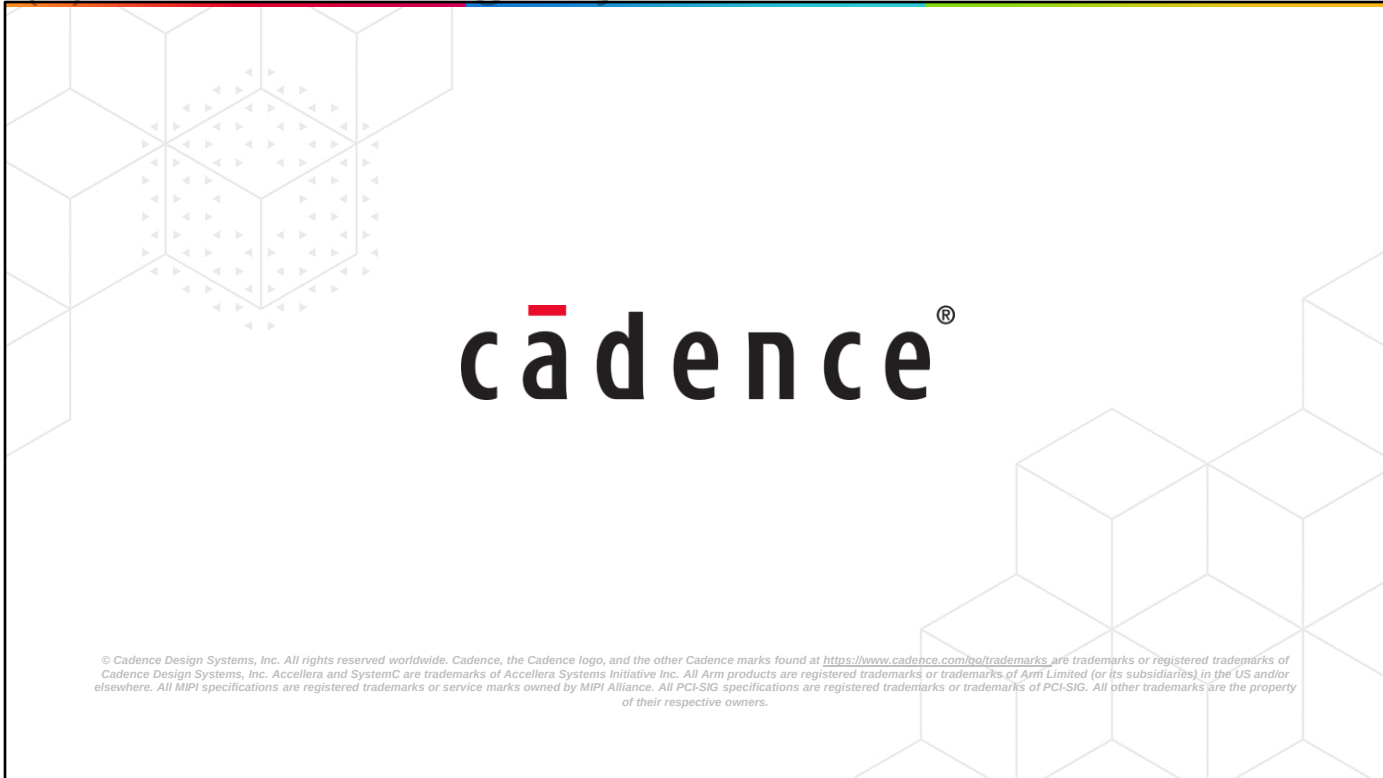
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Summary: Feedback

- General feedback structure
- Effects of negative feedback
 - Gain sensitivity
 - Nonlinear distortion
 - Frequency response
- Feedback topologies – shunt-shunt, series-series, series-shunt, shunt-series
 - Identification of topology
 - Amplifier configuration
 - Derivation of open-loop gain, feedback factor, closed-loop gain, input and output impedances
 - Examples and solutions



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